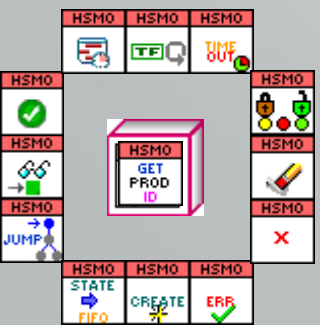
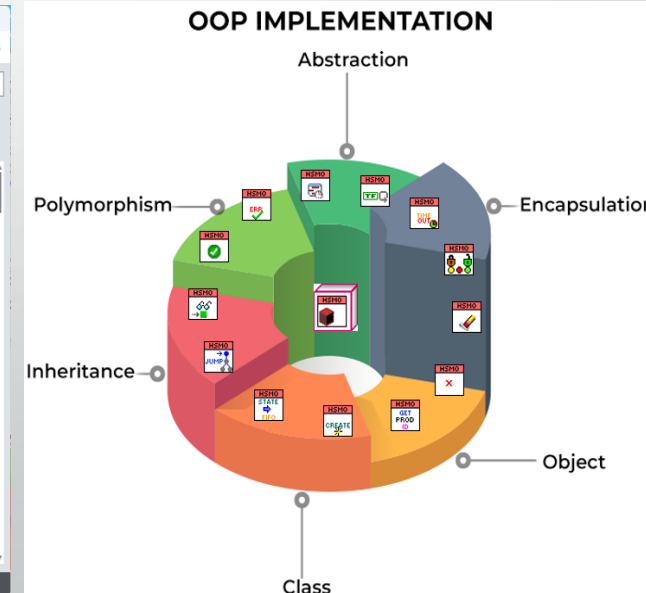
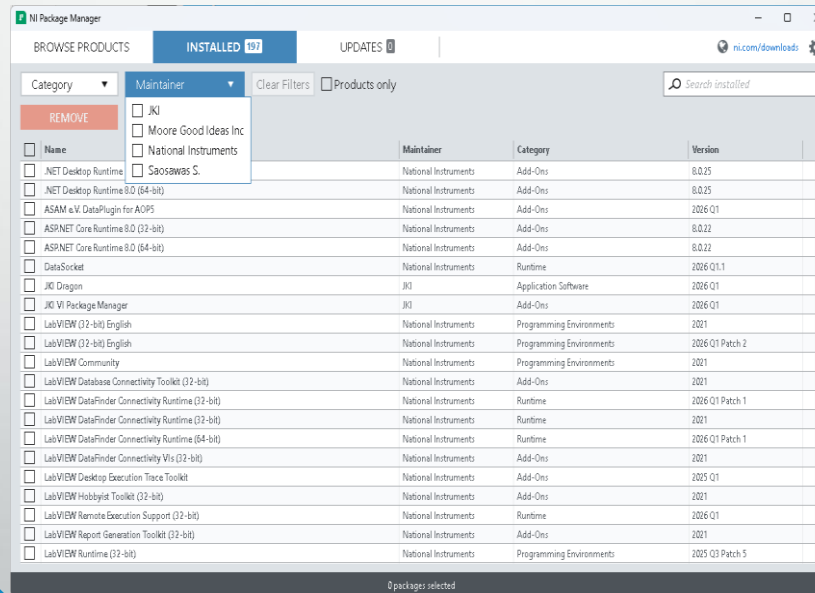
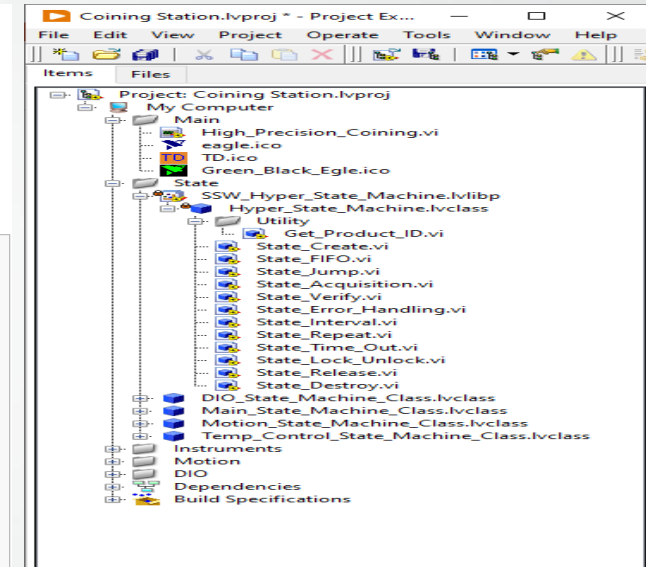
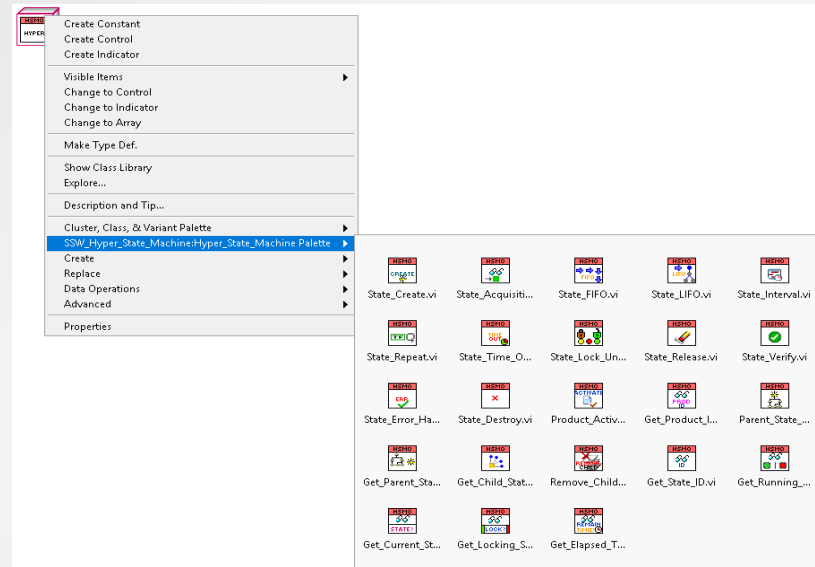


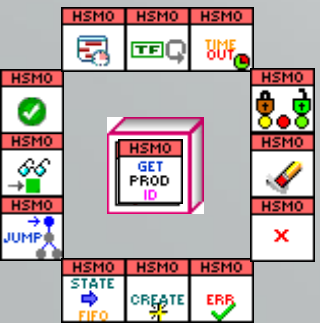
Hierarchical State Machine Object (HSMO)

- ❖ Background
- ❖ Framework Installation
- ❖ Framework Remove
- ❖ Object Oriented Programming
- ❖ Shortcut Menu & Class Member
- ❖ Design Pattern
- ❖ Implementation
- ❖ Benefit



Hierarchical State Machine Object (HSMO)

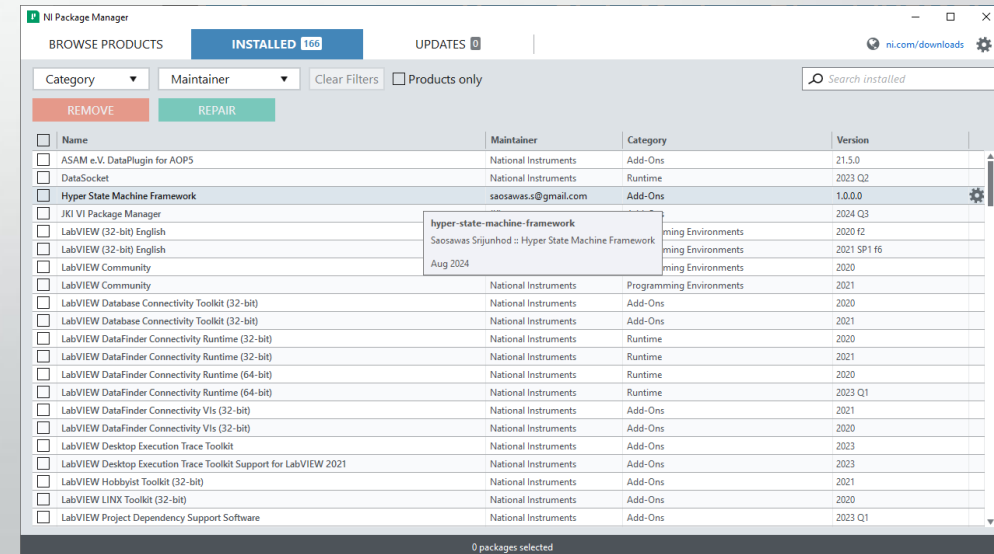
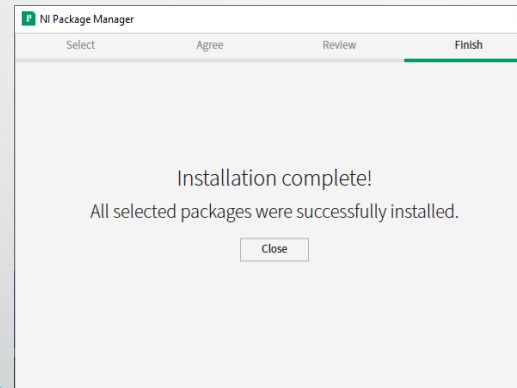
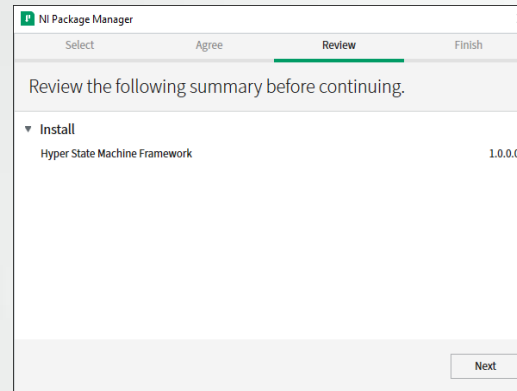
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Hierarchical State Machine Object (HSMO)

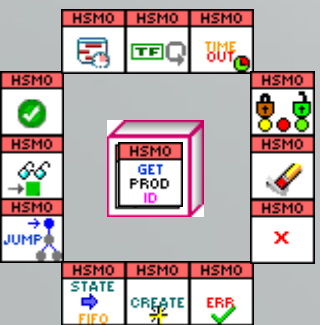
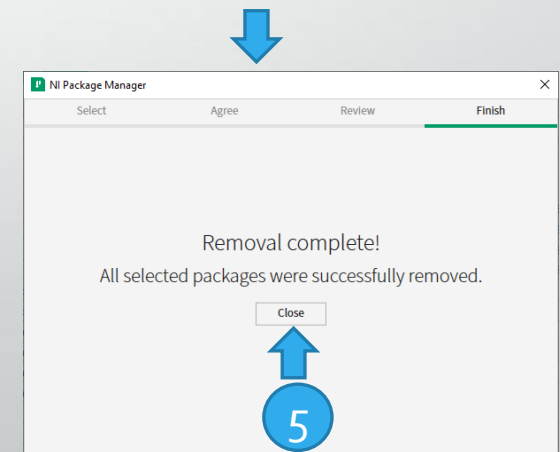
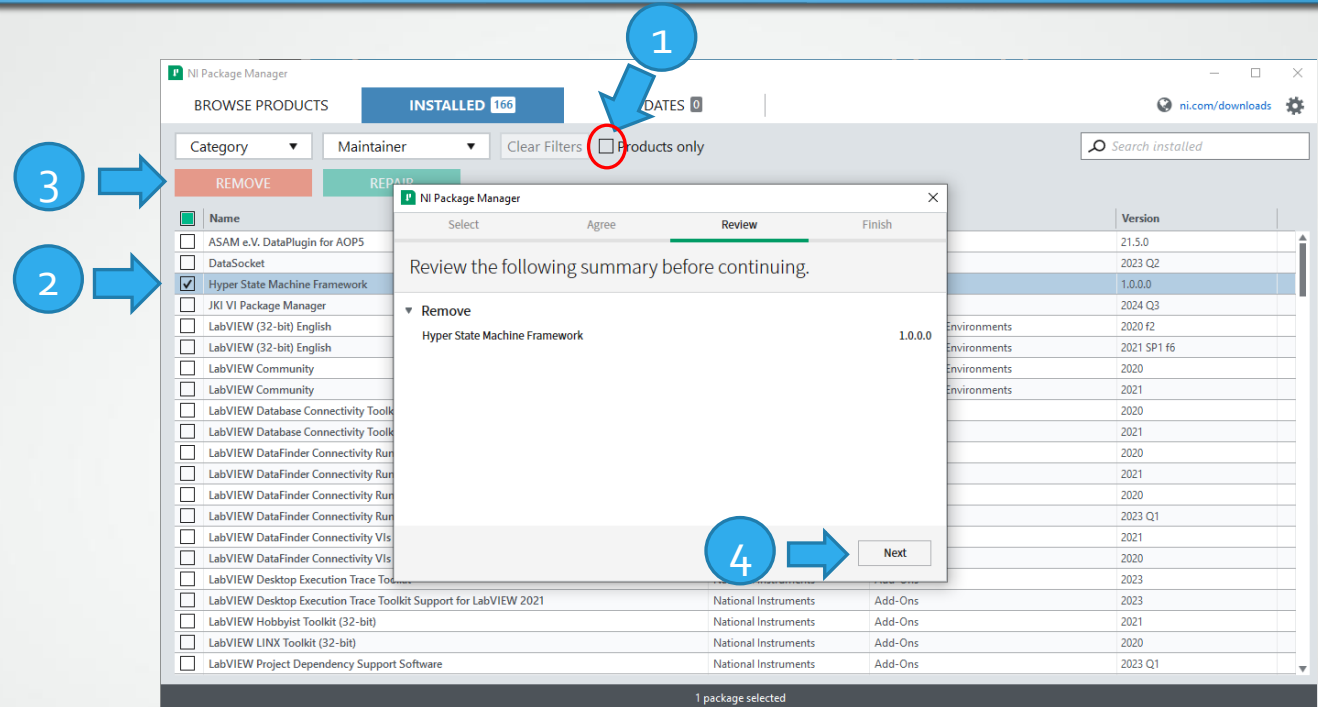
❖ Framework Installation

hyper-state-machine-framework_1.0.0-0_windows_all.nipkg



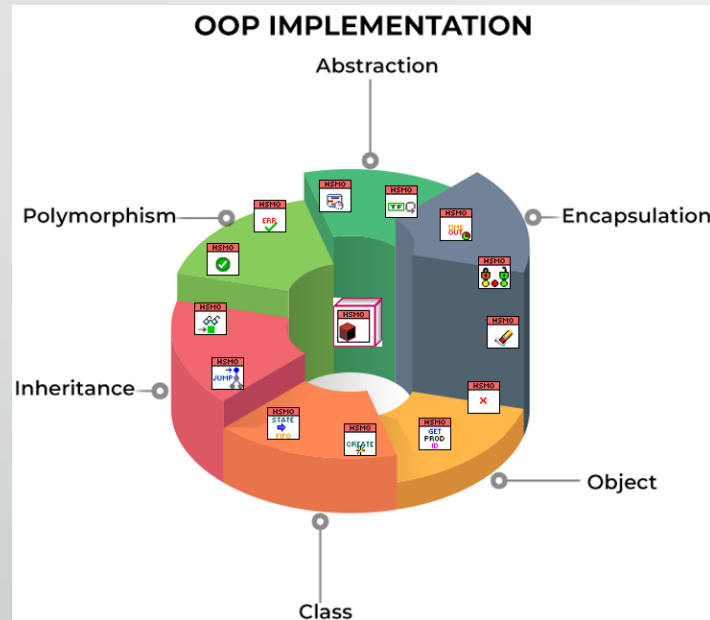
Hierarchical State Machine Object (HSMO)

Framework Remove



Hierarchical State Machine Object (HSMO)

❖ Object Oriented Programming (OOP)



❖ Abstraction :

Data abstraction is a design pattern in which data are visible only to semantically related functions, so as to prevent misuse. The success of data abstraction leads to frequent incorporation of data hiding as a design principle in object oriented and pure functional programming.

❖ Inheritance :

inheritance. This allows classes to be arranged in a hierarchy that represents "is-a-type-of" relationships. For example, class Employee might inherit from class Person. All the data and methods available to the parent class also appear in the child class with the same names. For example, class Person might define variables "first_name" and "last_name" with method "make_full_name()". These will also be available in class Employee, which might add the variables "position" and "salary". This technique allows easy re-use of the same procedures and data definitions, in addition to potentially mirroring real-world relationships in an intuitive way. Rather than utilizing database tables and programming subroutines, the developer utilizes objects the user may be more familiar with: objects from their application domain.

❖ Encapsulation :

Encapsulation prevents external code from being concerned with the internal workings of an object. This facilitates code refactoring, for example allowing the author of the class to change how objects of that class represent their data internally without changing any external code (as long as "public" method calls work the same way). It also encourages programmers to put all the code that is concerned with a certain set of data in the same class, which organizes it for easy comprehension by other programmers. Encapsulation is a technique that encourages decoupling.

❖ Polymorphism :

polymorphism – is when calling code can be independent of which class in the supported hierarchy it is operating on – the parent class or one of its descendants. Meanwhile, the same operation name among objects in an inheritance hierarchy may behave differently.

For example, objects of type Circle and Square are derived from a common class called Shape. The Draw function for each type of Shape implements what is necessary to draw itself while calling code can remain indifferent to the particular type of Shape being drawn.

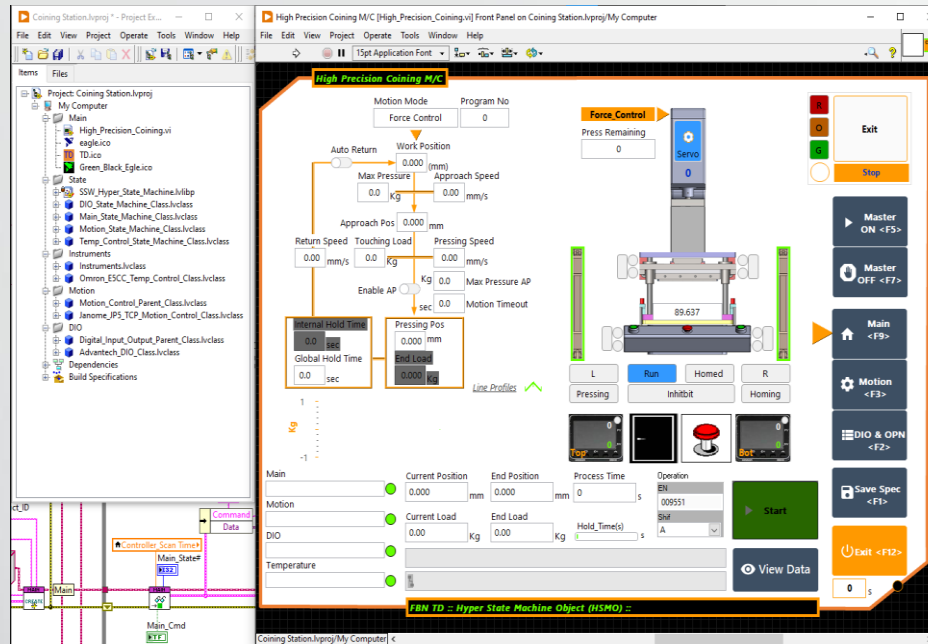
This is another type of abstraction that simplifies code external to the class hierarchy and enables strong separation of concerns.

❖ Object is an Instant of Class

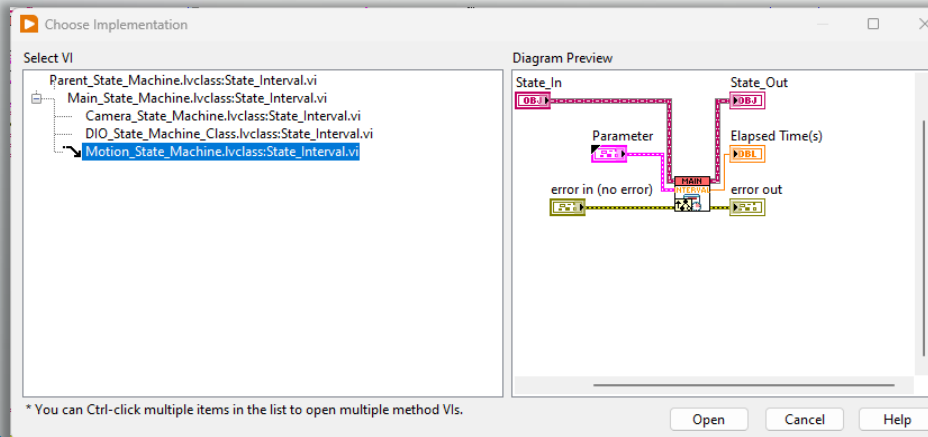
❖ In object-oriented programming, a class is a template definition of the methods and variables in a particular kind of object. Thus, an object is a specific instance of a class; it contains real values instead of variables. The class is one of the defining ideas of object-oriented programming.

Hierarchical State Machine Object (HSMO)

Project

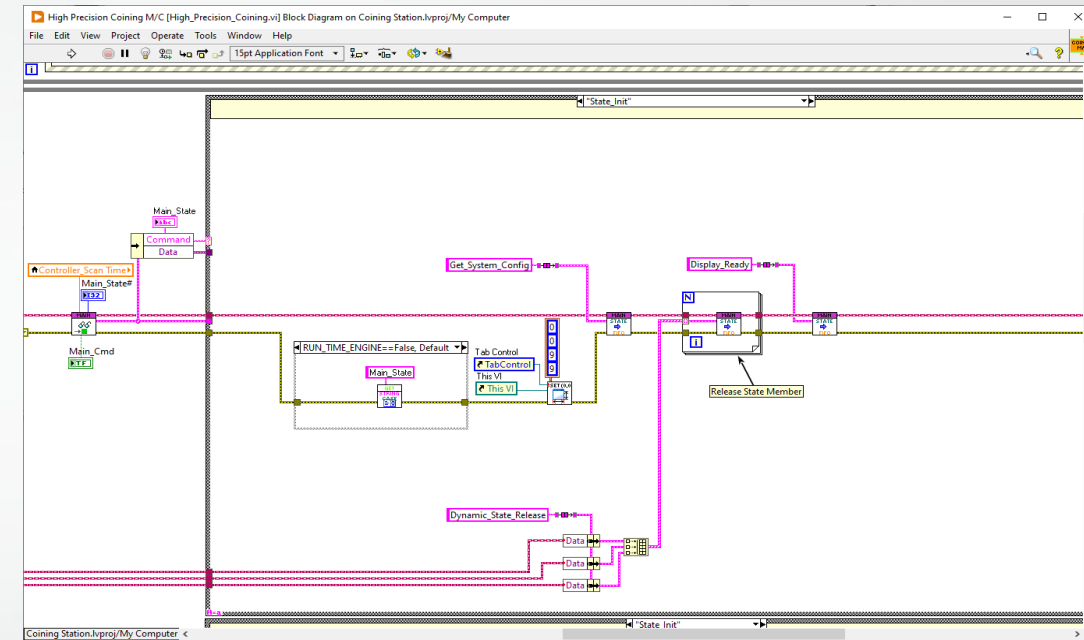


Dynamic Dispatching

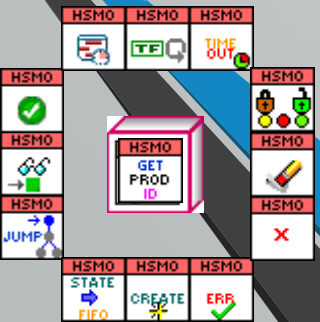
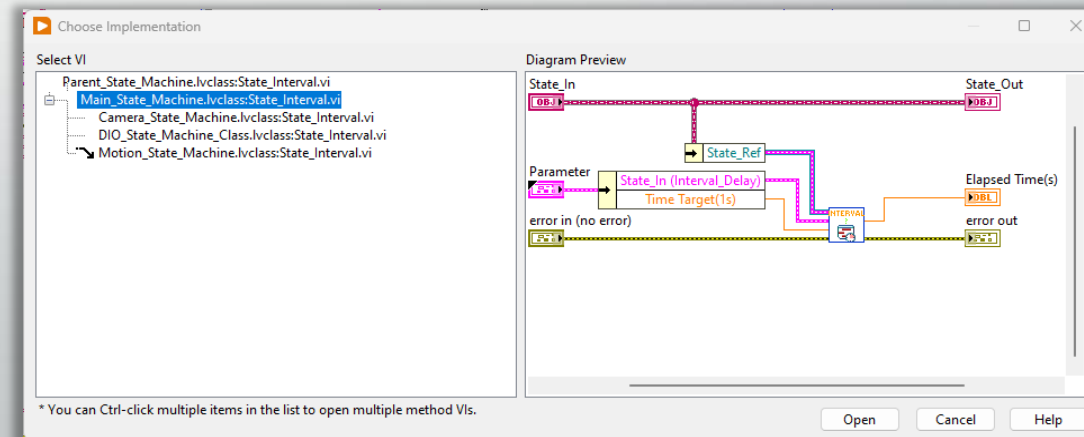


Coding

Implementation)



Method Overriding

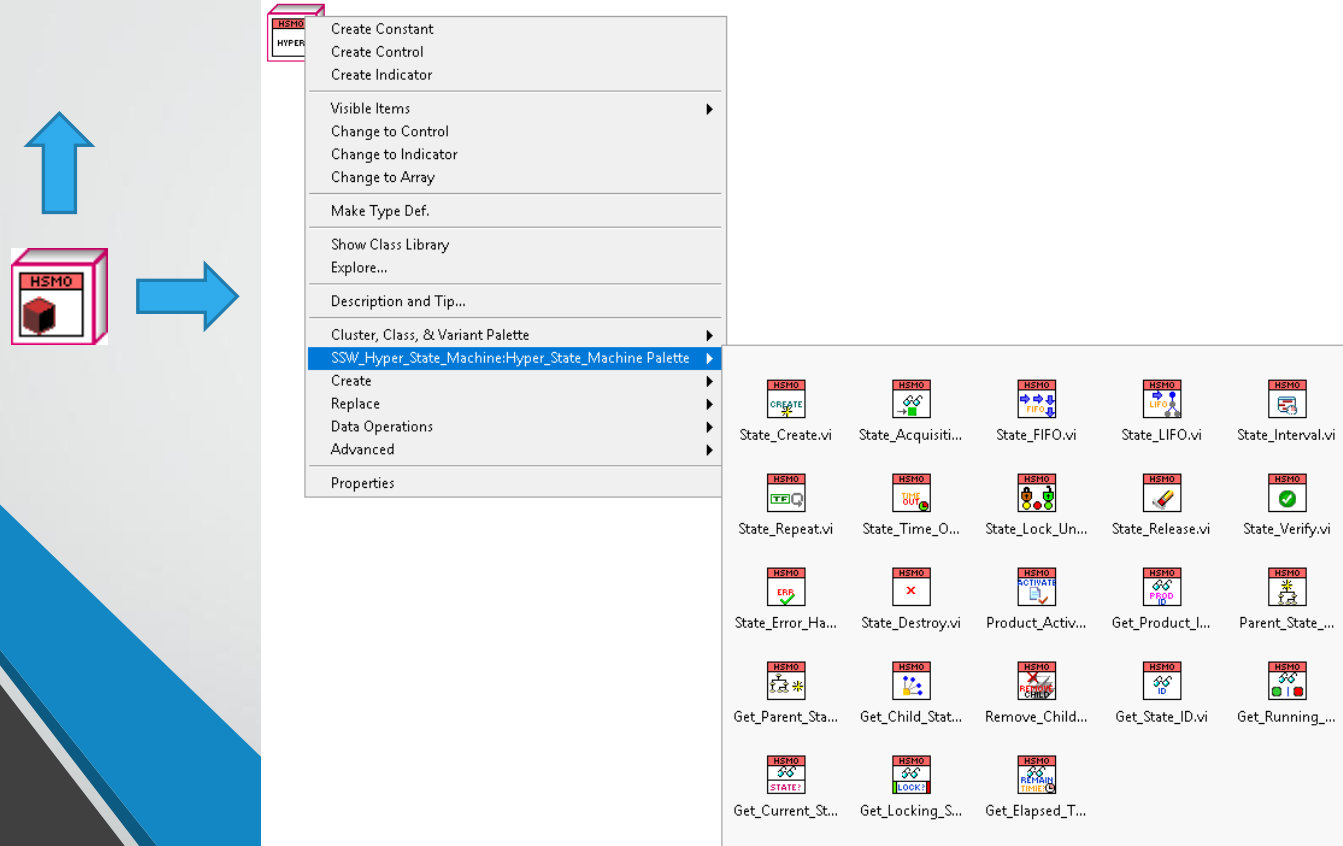


Hierarchical State Machine Object (HSMO)

SSW_Hyper_State_Machine.lvlib:Hyper_State_Machine.lvclass

No description available.

 SSW_Hyper_State_Machine.lvlib:Hyper_State_Machine.lvclass (SSW_Hyper_State_Machine.lvlib:Hyper_State_Machine)



❖ Shortcut Menu

❖ Class Member (23)

- Get_Product_ID
- Get_Running_State
- Get_Child_State
- Get_Paent_State
- Get_Current_State
- Get_Elapsed_Time
- Get_Locking_State
- Get_State_ID
- Parent_State_Change
- Product_Activate
- Remove_Child_State
- State_Create
- State_FIFO
- State_LIFO
- State_Acquisition
- State_Error_Handling
- State_Interval
- State_Repeat
- State_Time_Out
- State_Lock_Unlock
- State_Verify
- State_Release
- State_Destroy

Hierarchical State Machine Object (HSMO)

❖ Shortcut Menu

The image displays the HSMO shortcut menu and the SSW_Hyper_State_Machine:Hyper_State_Machine Palette. The menu is open, showing options for creating and managing HSMO objects. The palette contains various HSMO objects, each with a unique icon and name.

Shortcut Menu Options:

- Create Constant
- Create Control
- Create Indicator
- Visible Items
- Change to Control
- Change to Indicator
- Change to Array
- Make Type Def.
- Show Class Library
- Explore...
- Description and Tip...
- Cluster, Class, & Variant Palette
- SSW_Hyper_State_Machine:Hyper_State_Machine Palette**
- Create
- Replace
- Data Operations
- Advanced
- Properties

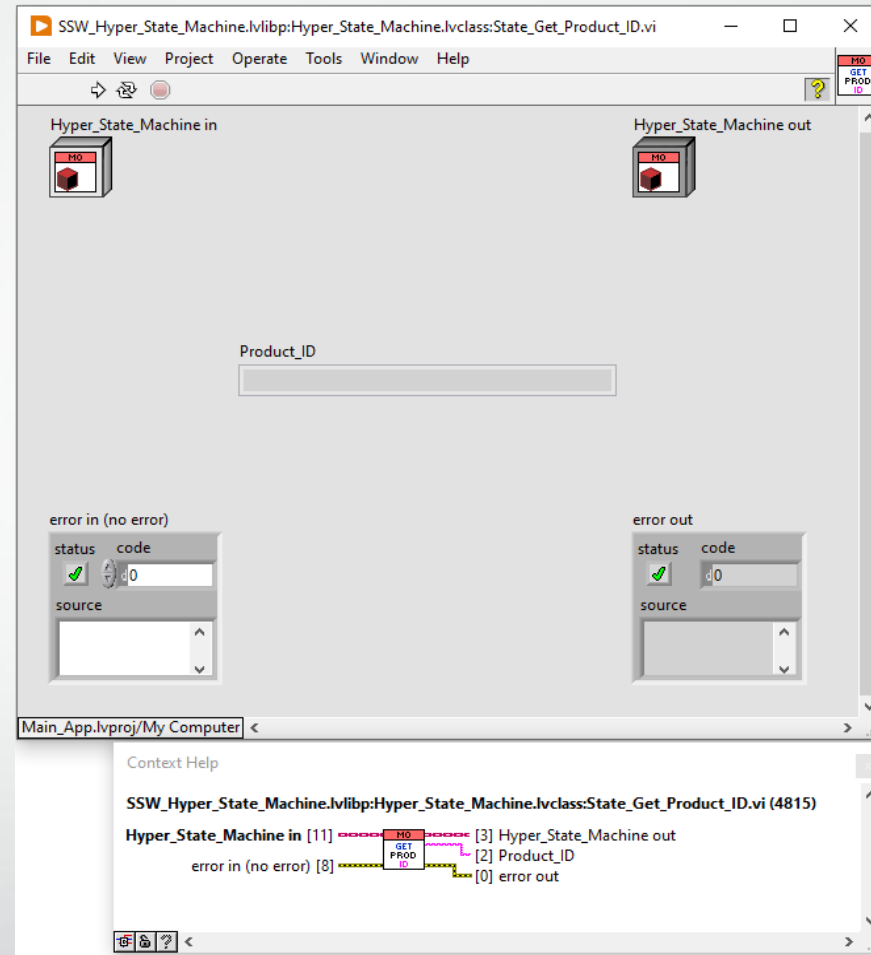
SSW_Hyper_State_Machine:Hyper_State_Machine Palette Objects:

Object Name	Object Name	Object Name	Object Name	Object Name
State_Create.vi	State_Acquisiti...	State_FIFO.vi	State_LIFO.vi	State_Interval.vi
State_Repeat.vi	State_Time_O...	State_Lock_Un...	State_Release.vi	State_Verify.vi
State_Error_Ha...	State_Destroy.vi	Product_Activ...	Get_Product_L...	Parent_State_...
Get_Parent_Sta...	Get_Child_Stat...	Remove_Child...	Get_State_ID.vi	Get_Running_...
Get_Current_St...	Get_Locking_S...	Get_Elapsed_T...		

Hierarchical State Machine Object (HSMO)

Class Member (23)

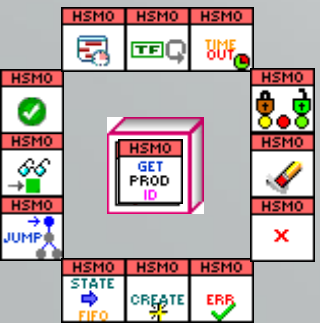
- [Get_Product_ID](#)
- State_Create
- State_FIFO
- State_LIFO
- State_Acquisition
- State_Error_Handling
- State_Interval
- State_Repeat
- State_Time_Out
- State_Lock_Unlock
- State_Verify
- State_Release
- State_Destroy
- Product_Activate
- Parent_State_Override
- Get_Parent_State
- Get_Child_State
- Remove_Child_State
- Get_State_ID
- Get_Running_State
- Get_Current_State
- Get_Locking_State
- Get_Elapsed_Time



Method :

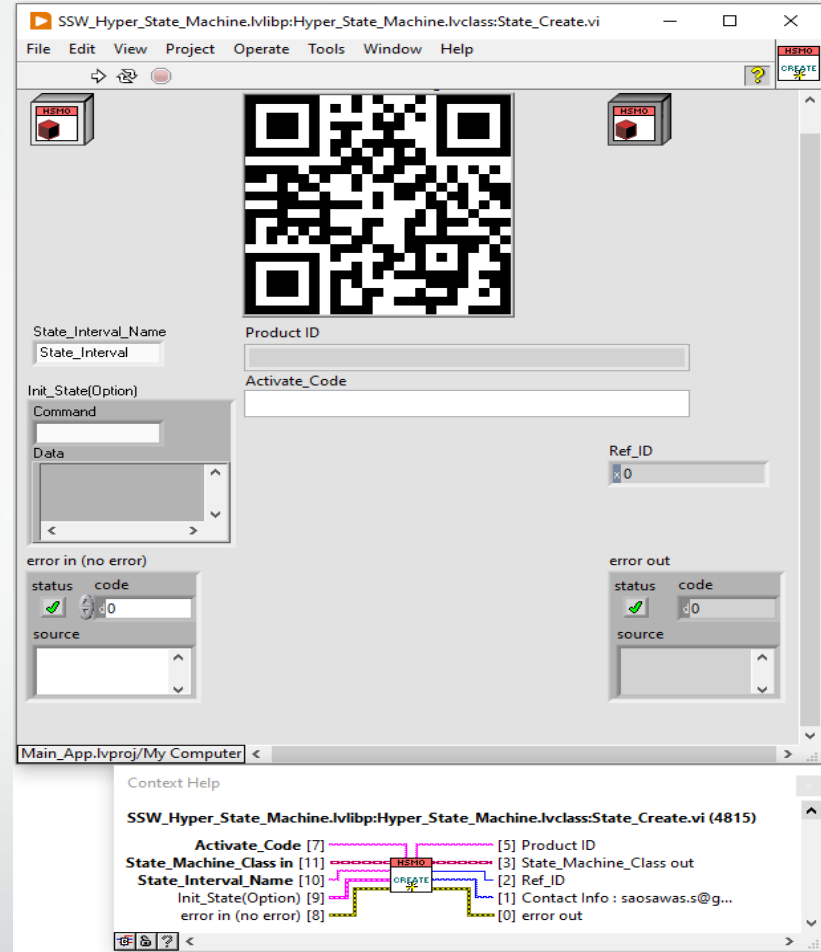


Get HSMO Product ID for License Activate in State_Create Method

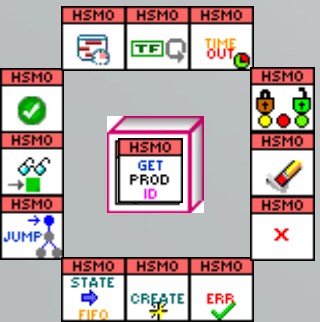


Hierarchical State Machine Object (HSMO)

- [Get_Product_ID](#)
- [State_Create](#)
- [State_FIFO](#)
- [State_LIFO](#)
- [State_Acquisition](#)
- [State_Error_Handling](#)
- [State_Interval](#)
- [State_Repeat](#)
- [State_Time_Out](#)
- [State_Lock_Unlock](#)
- [State_Verify](#)
- [State_Release](#)
- [State_Destroy](#)
- [Product_Activate](#)
- [Parent_State_Override](#)
- [Get_Parent_State](#)
- [Get_Child_State](#)
- [Remove_Child_State](#)
- [Get_State_ID](#)
- [Get_Running_State](#)
- [Get_Current_State](#)
- [Get_Locking_State](#)
- [Get_Elapsed_Time](#)



Method :

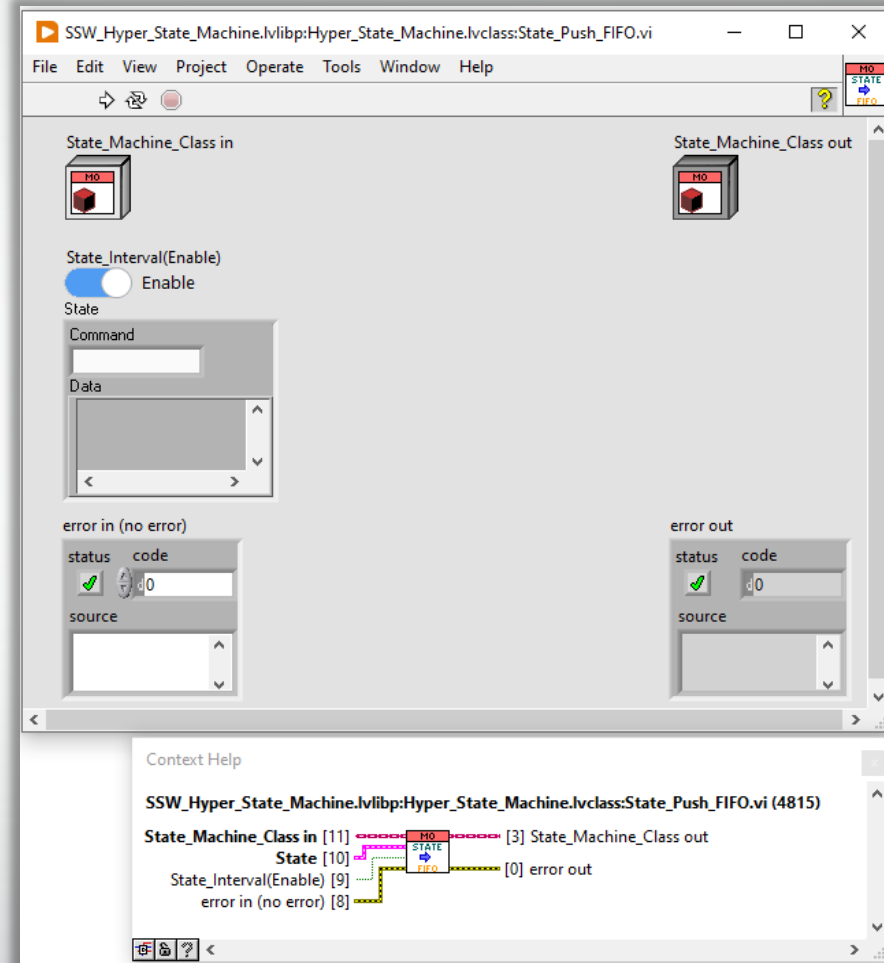
 Create HSMO

Hierarchical State Machine Object (HSMO)



Class Member (23)

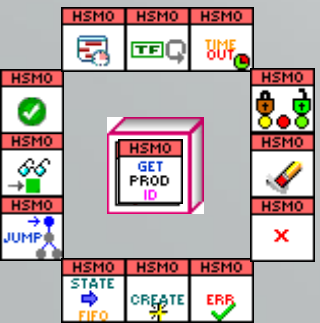
- Get_Product_ID
- State_Create
- State_FIFO
- State_LIFO
- State_Acquisition
- State_Error_Handling
- State_Interval
- State_Repeat
- State_Time_Out
- State_Lock_Unlock
- State_Verify
- State_Release
- State_Destroy
- Product_Activate
- Parent_State_Override
- Get_Parent_State
- Get_Child_State
- Remove_Child_State
- Get_State_ID
- Get_Running_State
- Get_Current_State
- Get_Locking_State
- Get_Elapsed_Time



Method :

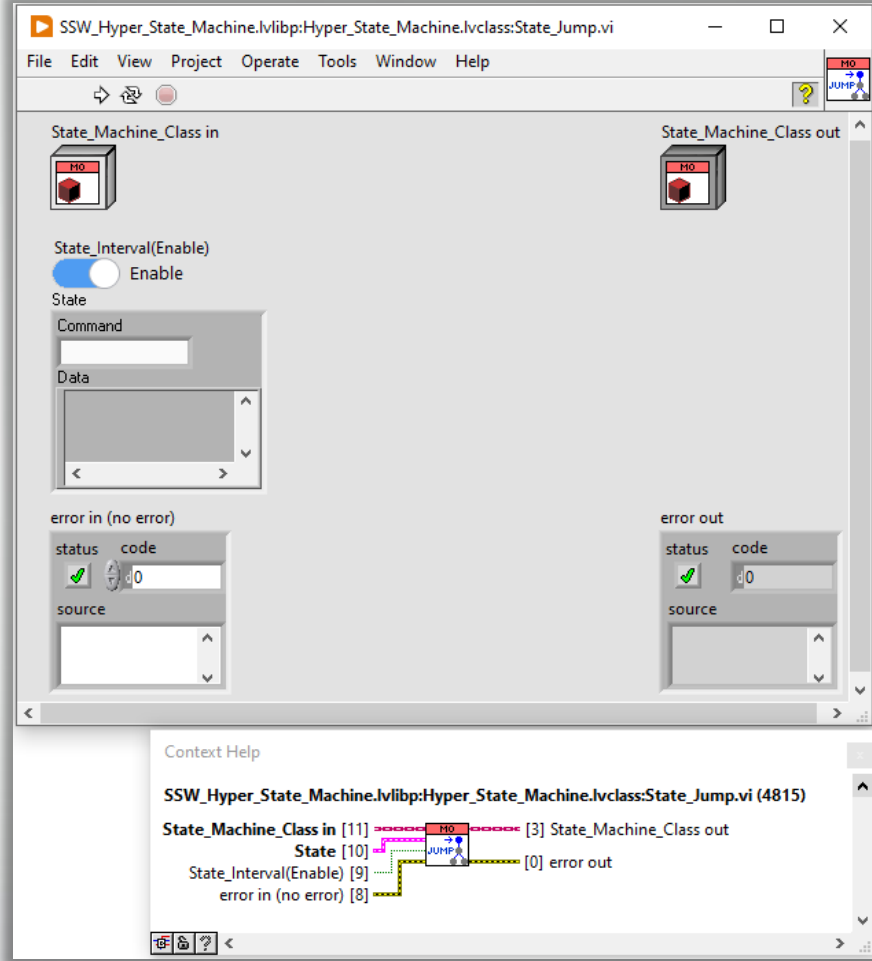


Append HSMO as FIFO (First In First Out)

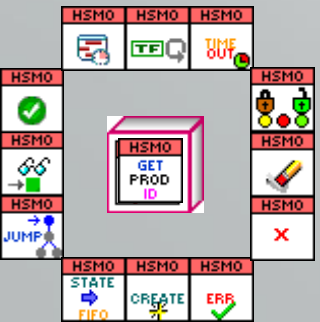


Hierarchical State Machine Object (HSMO)

- Get_Product_ID
- State_Create
- State_FIFO
- State_LIFO
- State_Acquisition
- State_Error_Handling
- State_Interval
- State_Repeat
- State_Time_Out
- State_Lock_Unlock
- State_Verify
- State_Release
- State_Destroy
- Product_Activate
- Parent_State_Override
- Get_Parent_State
- Get_Child_State
- Remove_Child_State
- Get_State_ID
- Get_Running_State
- Get_Current_State
- Get_Locking_State
- Get_Elapsed_Time

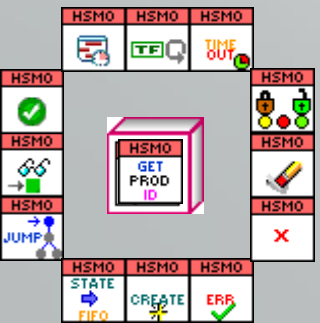
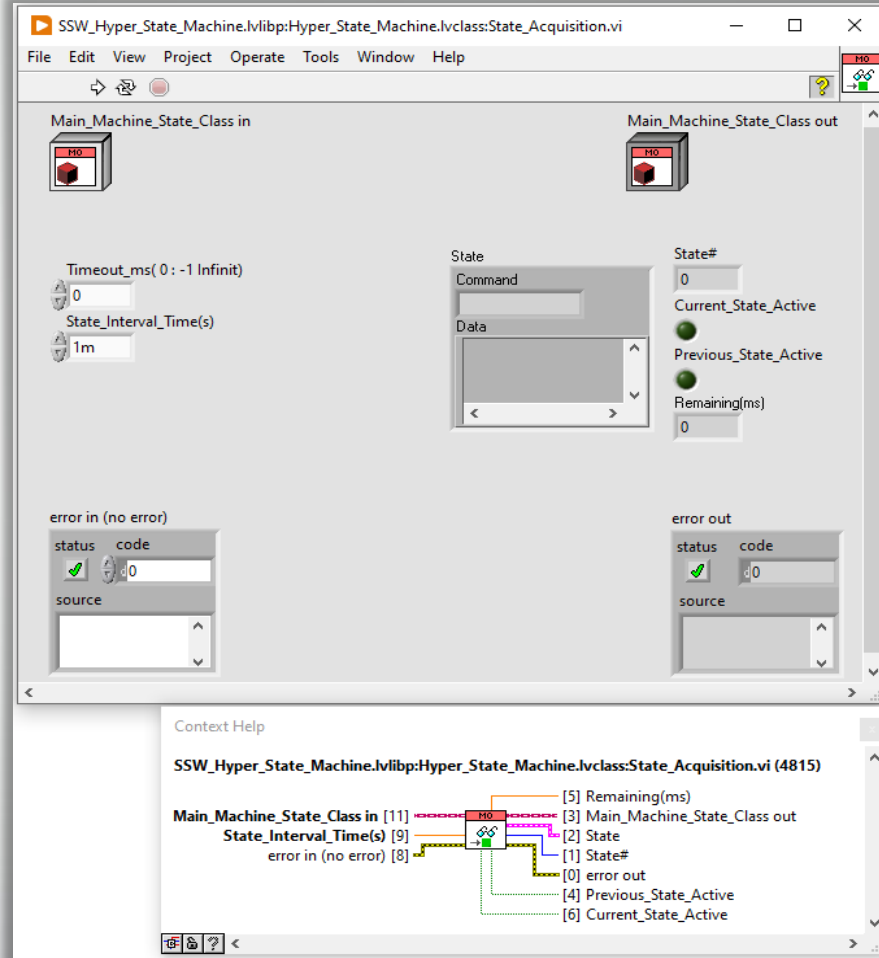


➤ Insert HSMO as Front most (Last In First Out)



Hierarchical State Machine Object (HSMO)

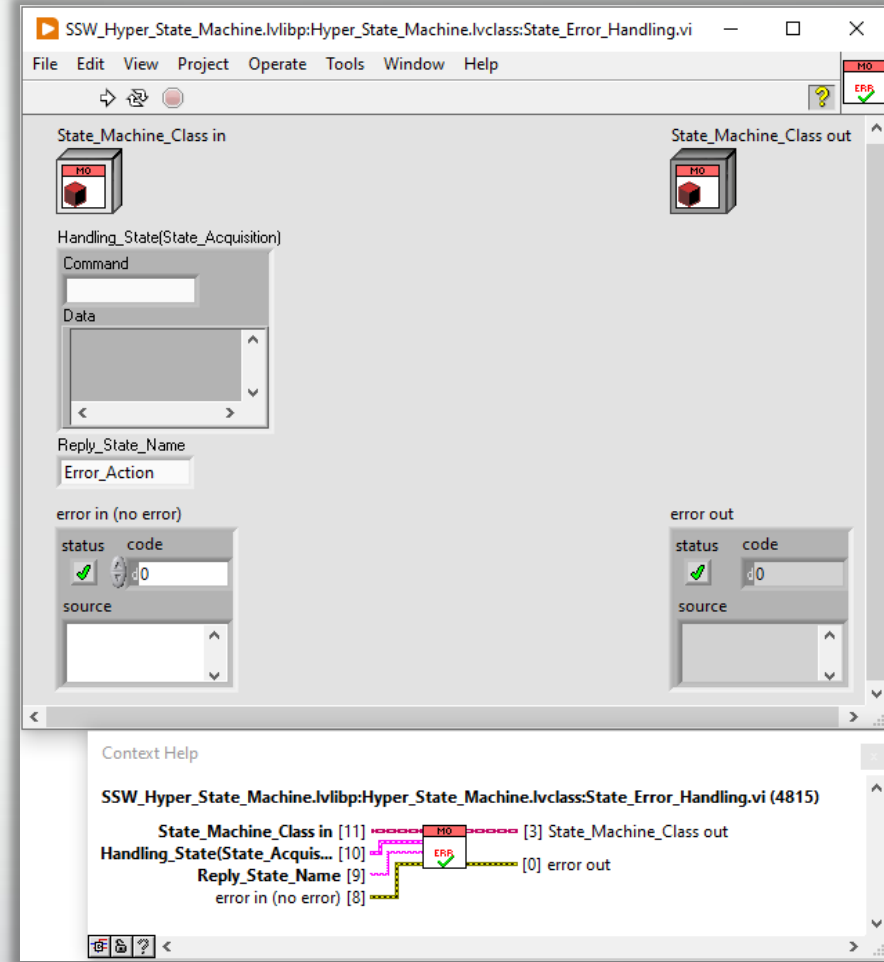
- Get_Product_ID
- State_Create
- State_FIFO
- State_LIFO
- State_Acquisition
- State_Error_Handling
- State_Interval
- State_Repeat
- State_Time_Out
- State_Lock_Unlock
- State_Verify
- State_Release
- State_Destroy
- Product_Activate
- Parent_State_Override
- Get_Parent_State
- Get_Child_State
- Remove_Child_State
- Get_State_ID
- Get_Running_State
- Get_Current_State
- Get_Locking_State
- Get_Elapsed_Time



Hierarchical State Machine Object (HSMO)

❖ Class Member (23)

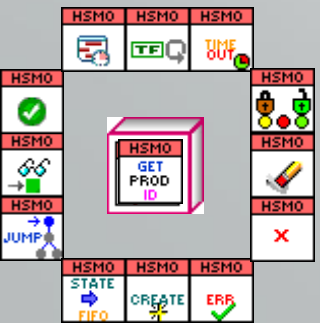
- Get_Product_ID
- State_Create
- State_FIFO
- State_LIFO
- State_Acquisition
- State_Error_Handling
- State_Interval
- State_Repeat
- State_Time_Out
- State_Lock_Unlock
- State_Verify
- State_Release
- State_Destroy
- Product_Activate
- Parent_State_Override
- Get_Parent_State
- Get_Child_State
- Remove_Child_State
- Get_State_ID
- Get_Running_State
- Get_Current_State
- Get_Locking_State
- Get_Elapsed_Time



❖ Method :



Error Handling of HSMO Member Class and Un define Method Detection

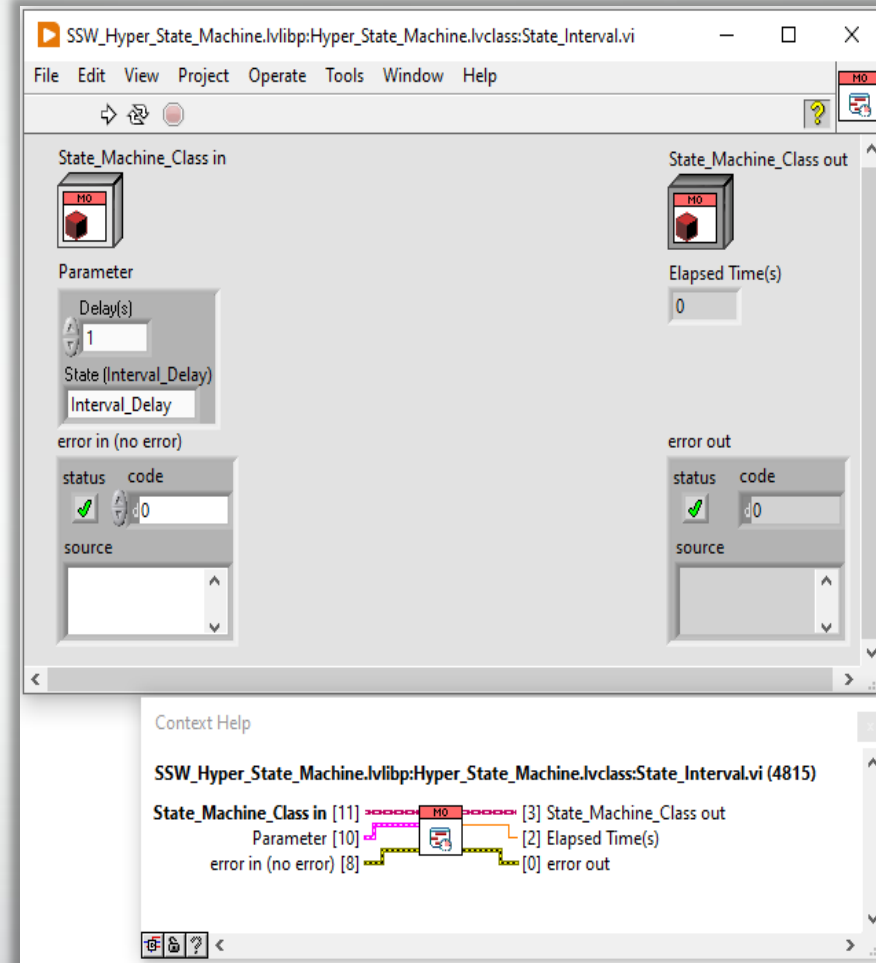


Hierarchical State Machine Object (HSMO)



Class Member (23)

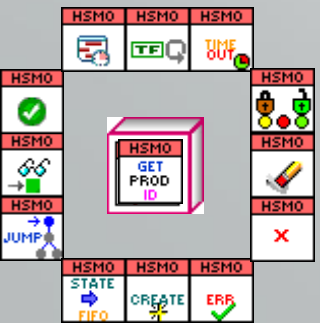
- Get_Product_ID
- State_Create
- State_FIFO
- State_LIFO
- State_Acquisition
- State_Error_Handling
- State_Interval
- State_Repeat
- State_Time_Out
- State_Lock_Unlock
- State_Verify
- State_Release
- State_Destroy
- Product_Activate
- Parent_State_Override
- Get_Parent_State
- Get_Child_State
- Remove_Child_State
- Get_State_ID
- Get_Running_State
- Get_Current_State
- Get_Locking_State
- Get_Elapsed_Time



Method :

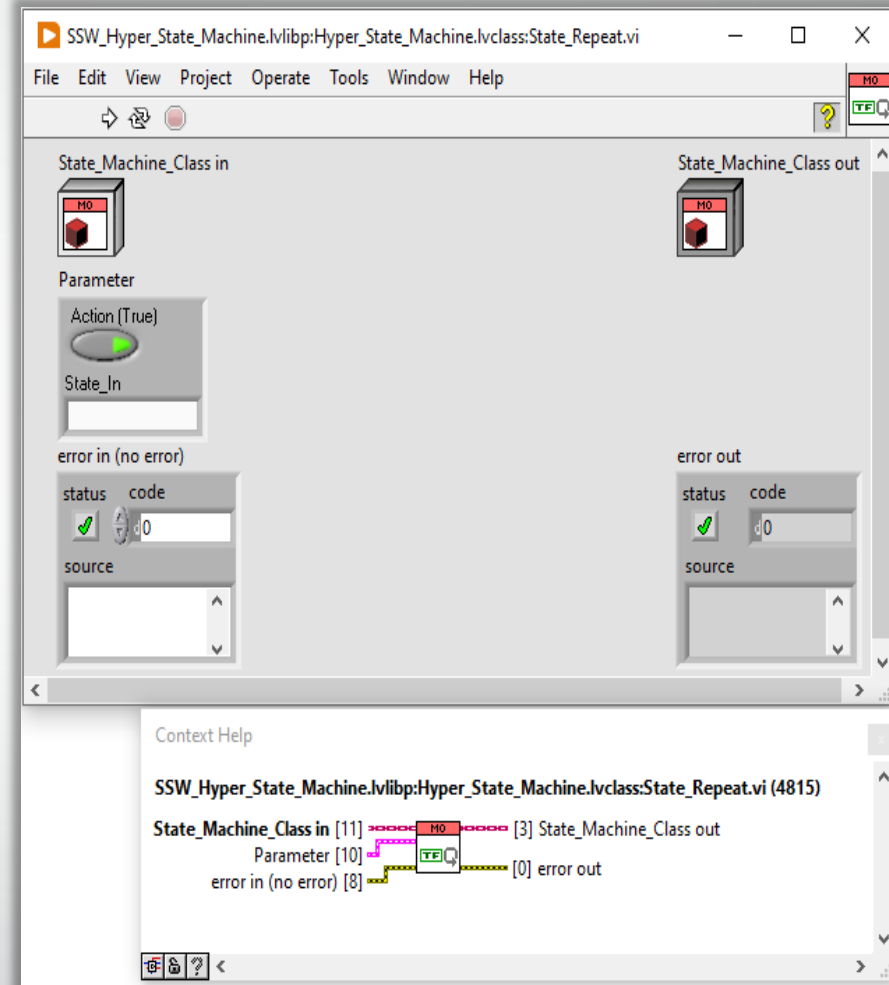


Repeat HSMO Target State with Specify Timing Target



Hierarchical State Machine Object (HSMO)

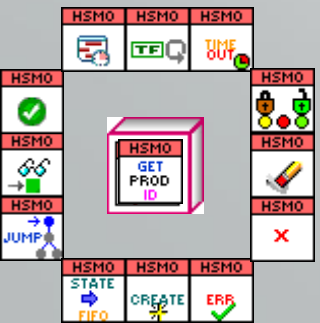
- ❖ Class Member (23)
- Get_Product_ID
 - State_Create
 - State_FIFO
 - State_LIFO
 - State_Acquisition
 - State_Error_Handling
 - State_Interval
 - State_Repeat
 - State_Time_Out
 - State_Lock_Unlock
 - State_Verify
 - State_Release
 - State_Destroy
 - Product_Activate
 - Parent_State_Override
 - Get_Parent_State
 - Get_Child_State
 - Remove_Child_State
 - Get_State_ID
 - Get_Running_State
 - Get_Current_State
 - Get_Locking_State
 - Get_Elapsed_Time



Method :



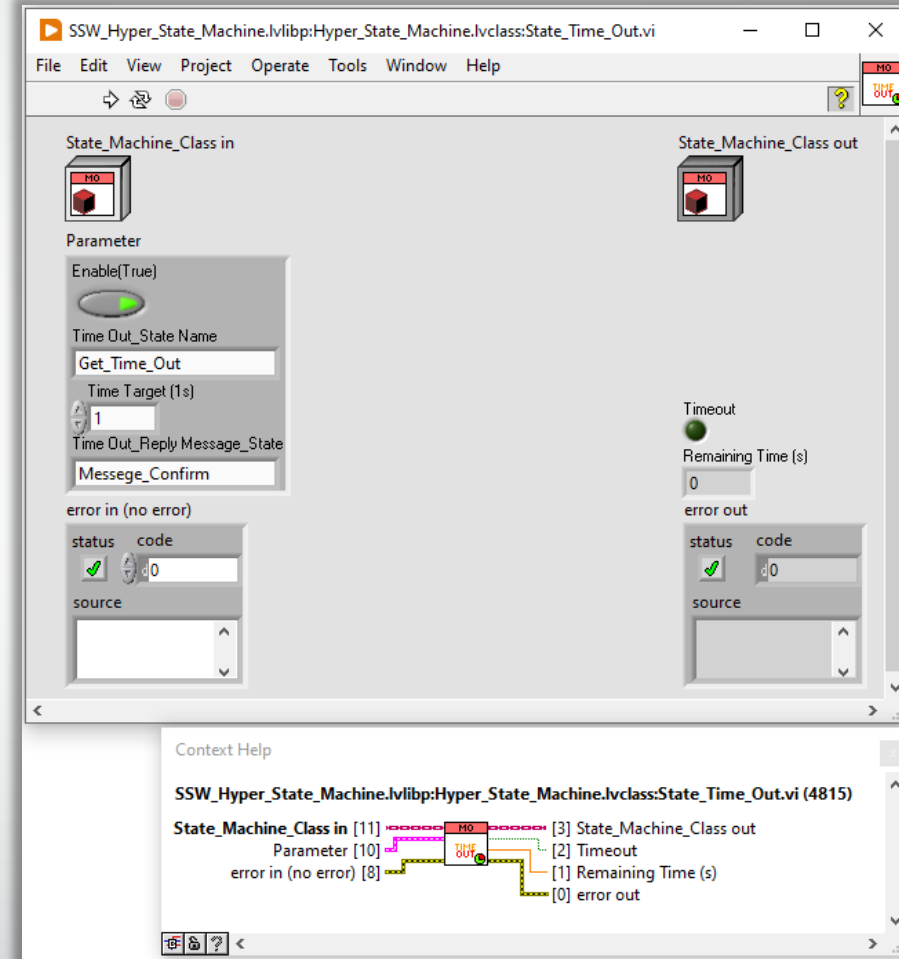
Repeat HSMO Target State with Condition of Flag (True/False)



Hierarchical State Machine Object (HSMO)

Class Member (23)

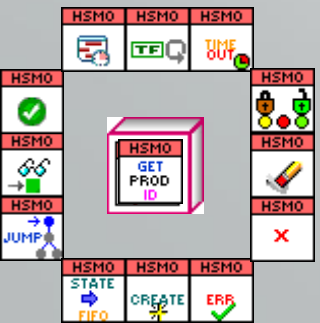
- Get_Product_ID
- State_Create
- State_FIFO
- State_LIFO
- State_Acquisition
- State_Error_Handling
- State_Interval
- State_Repeat
- State_Time_Out
- State_Lock_Unlock
- State_Verify
- State_Release
- State_Destroy
- Product_Activate
- Parent_State_Override
- Get_Parent_State
- Get_Child_State
- Remove_Child_State
- Get_State_ID
- Get_Running_State
- Get_Current_State
- Get_Locking_State
- Get_Elapsed_Time



Method :



Repeat & Monitoring HSMO Target State with Timeout Target and Condition of Flag (True/False) , if Target State is Timeout Active the Destination Action is activate

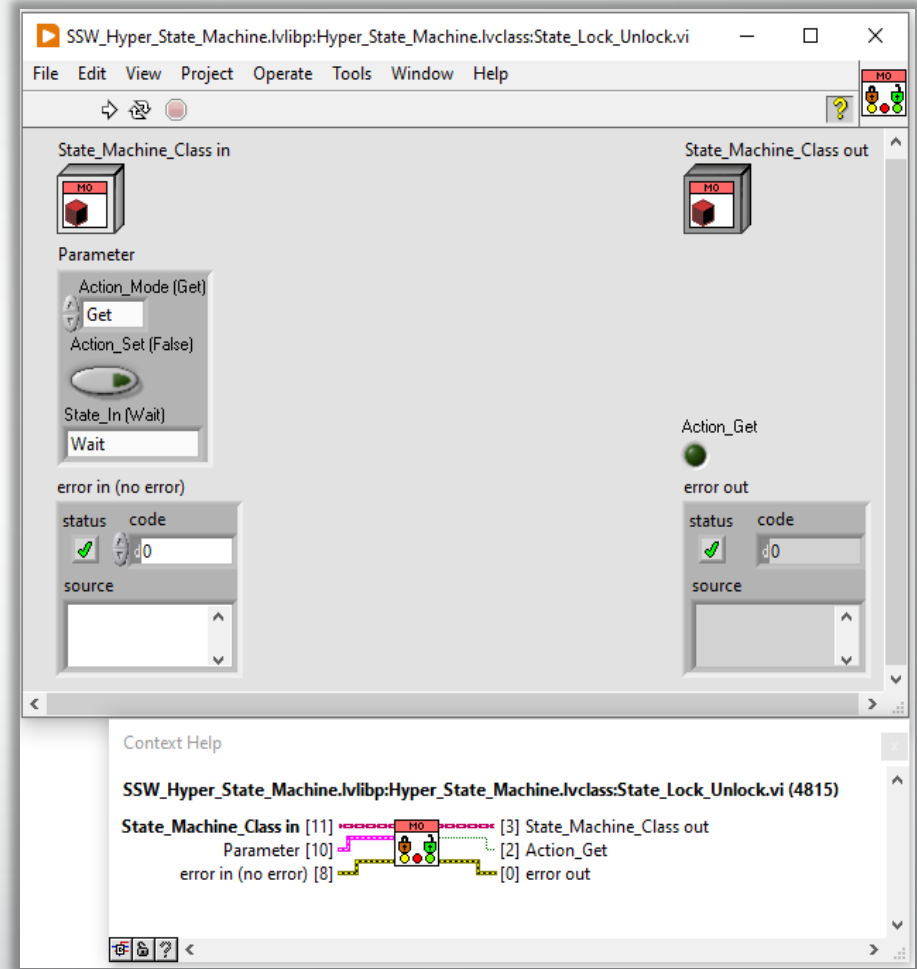


Hierarchical State Machine Object (HSMO)



Class Member (23)

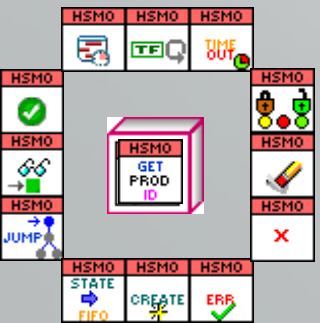
- Get_Product_ID
- State_Create
- State_FIFO
- State_LIFO
- State_Acquisition
- State_Error_Handling
- State_Interval
- State_Repeat
- State_Time_Out
- State_Lock_Unlock
- State_Verify
- State_Release
- State_Destroy
- Product_Activate
- Parent_State_Override
- Get_Parent_State
- Get_Child_State
- Remove_Child_State
- Get_State_ID
- Get_Running_State
- Get_Current_State
- Get_Locking_State
- Get_Elapsed_Time



Method :



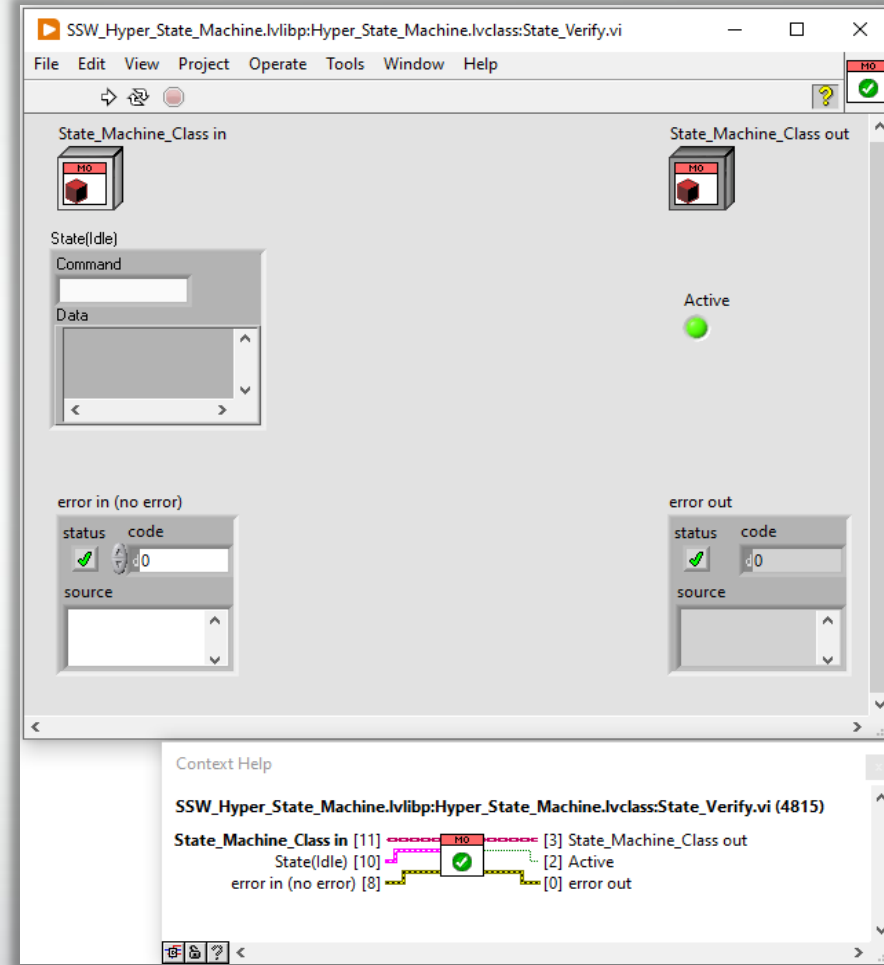
Lock or Unlock HSMO Target State, If Current State is Locking it's Waiting until the Unlock Command active



Hierarchical State Machine Object (HSMO)

❖ Class Member (23)

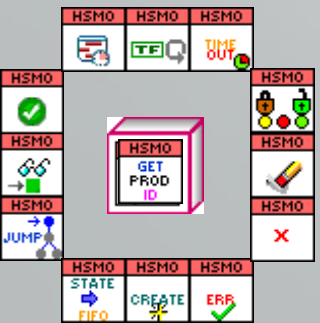
- Get_Product_ID
- State_Create
- State_FIFO
- State_LIFO
- State_Acquisition
- State_Error_Handling
- State_Interval
- State_Repeat
- State_Time_Out
- State_Lock_Unlock
- State_Verify
- State_Release
- State_Destroy
- Product_Activate
- Parent_State_Override
- Get_Parent_State
- Get_Child_State
- Remove_Child_State
- Get_State_ID
- Get_Running_State
- Get_Current_State
- Get_Locking_State
- Get_Elapsed_Time



Method :



HSMO Self detection of un define State or Action to prevent malfunction

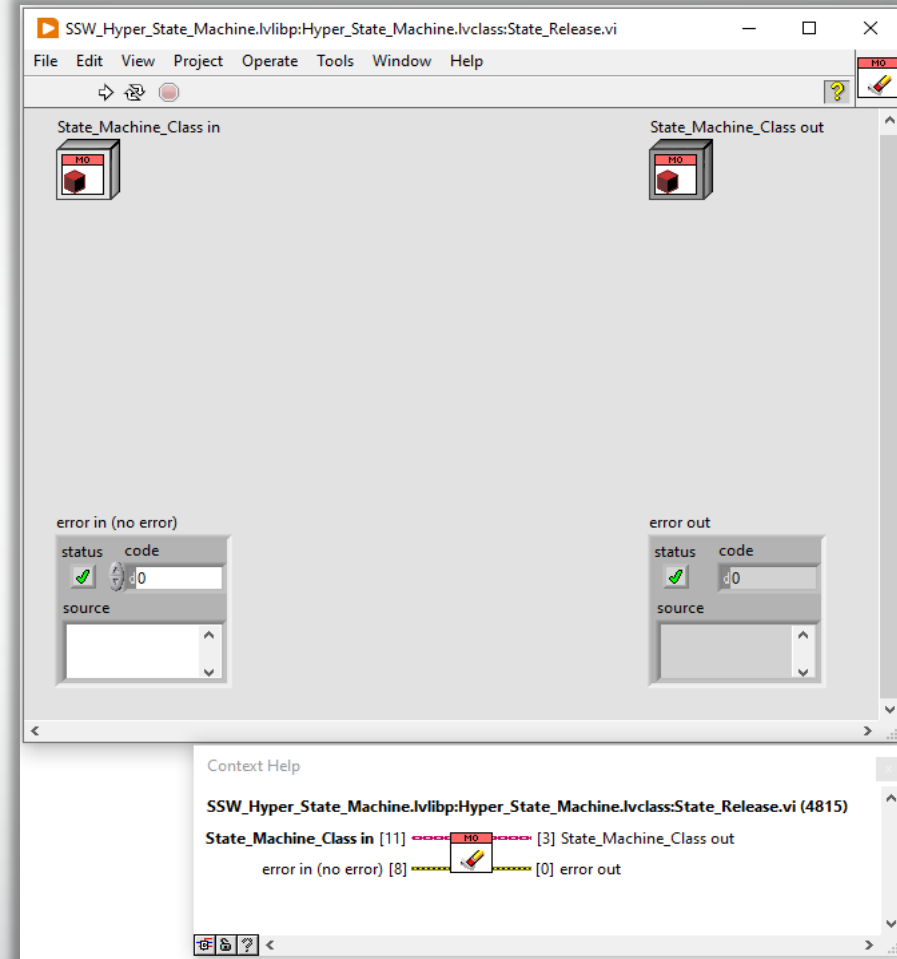


Hierarchical State Machine Object (HSMO)



Class Member (23)

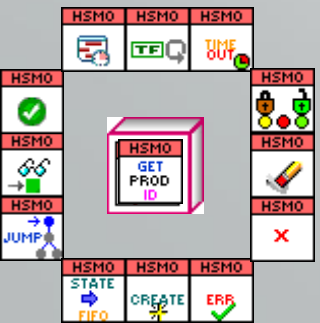
- Get_Product_ID
- State_Create
- State_FIFO
- State_LIFO
- State_Acquisition
- State_Error_Handling
- State_Interval
- State_Repeat
- State_Time_Out
- State_Lock_Unlock
- State_Verify
- State_Release
- State_Destroy
- Product_Activate
- Parent_State_Override
- Get_Parent_State
- Get_Child_State
- Remove_Child_State
- Get_State_ID
- Get_Running_State
- Get_Current_State
- Get_Locking_State
- Get_Elapsed_Time



Method :



HSMO Release or Buffer Flushing

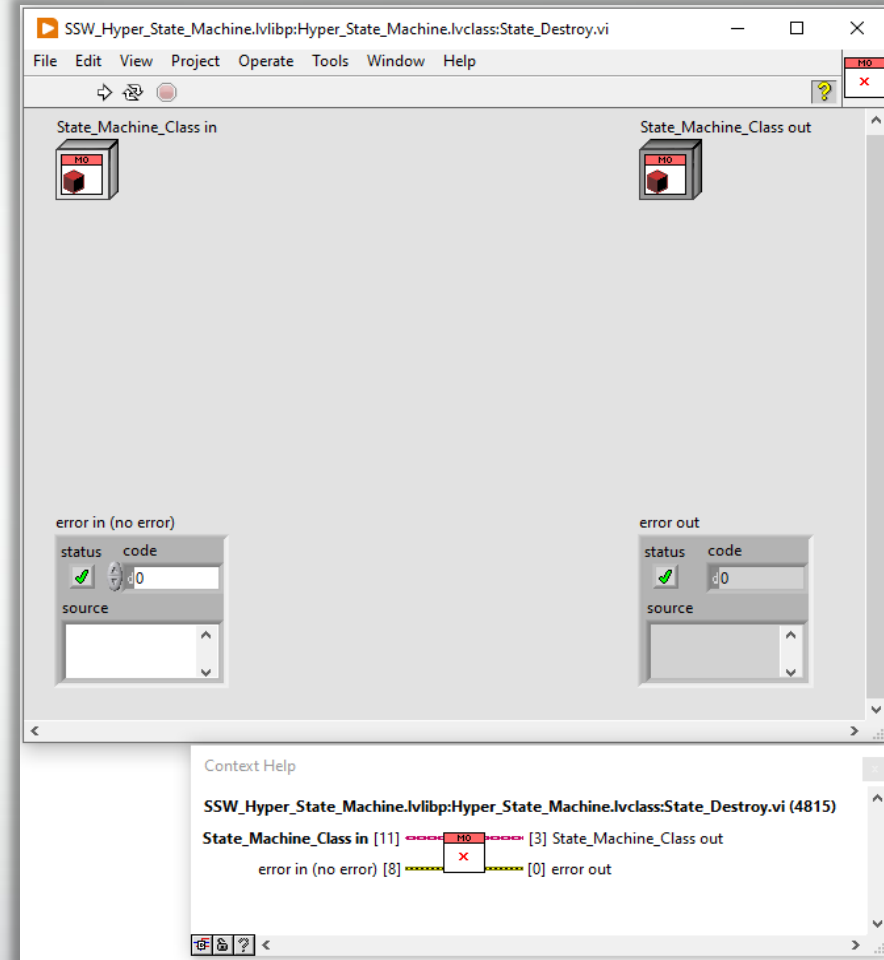


Hierarchical State Machine Object (HSMO)



Class Member (23)

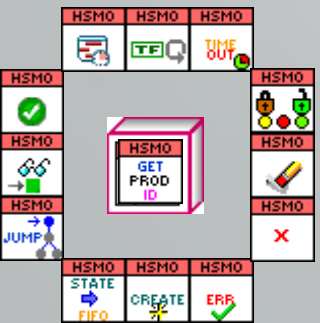
- Get_Product_ID
- State_Create
- State_FIFO
- State_LIFO
- State_Acquisition
- State_Error_Handling
- State_Interval
- State_Repeat
- State_Time_Out
- State_Lock_Unlock
- State_Verify
- State_Release
- State_Destroy
- Product_Activate
- Parent_State_Override
- Get_Parent_State
- Get_Child_State
- Remove_Child_State
- Get_State_ID
- Get_Running_State
- Get_Current_State
- Get_Locking_State
- Get_Elapsed_Time



Method :

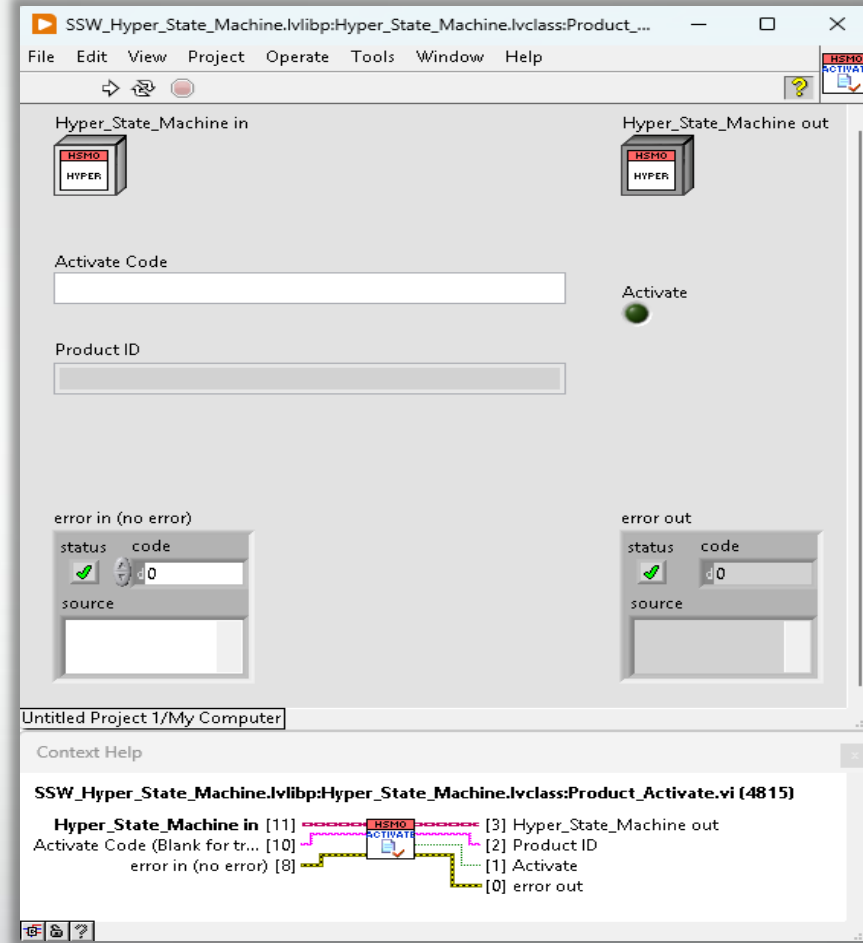
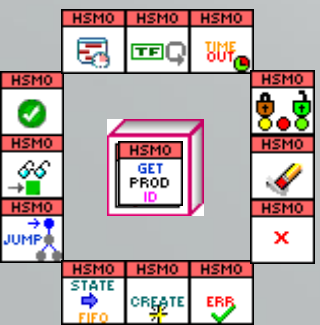


Delete HSMO



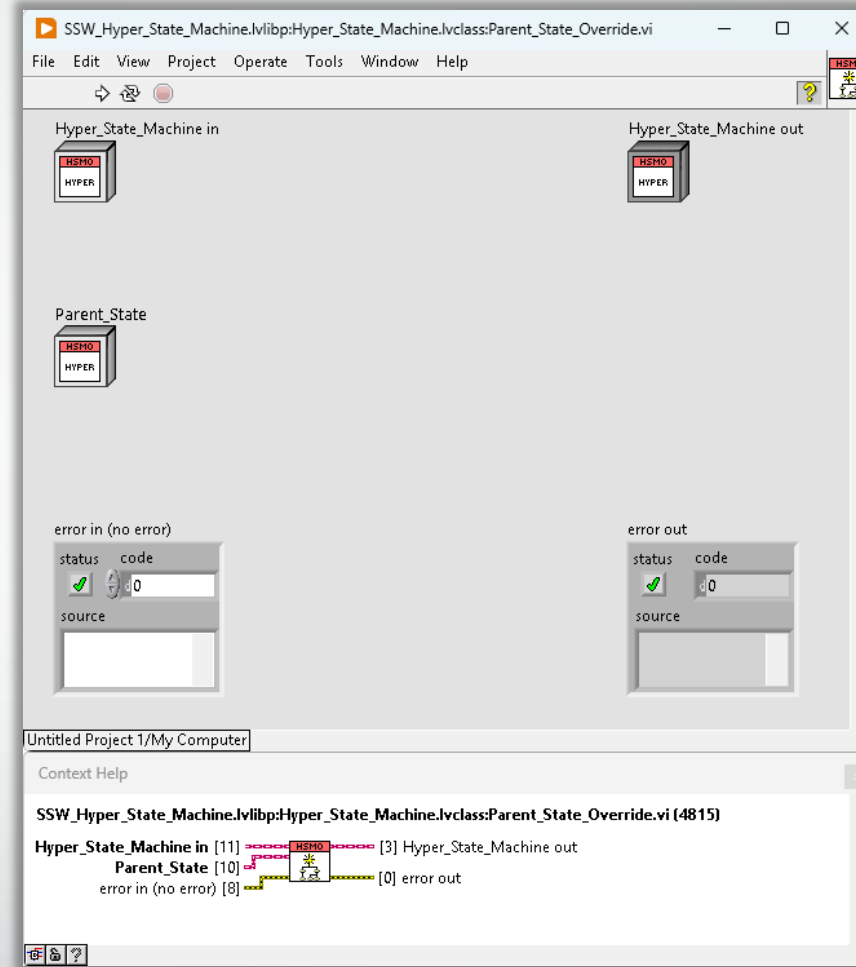
Hierarchical State Machine Object (HSMO)

- Get_Product_ID
- State_Create
- State_FIFO
- State_LIFO
- State_Acquisition
- State_Error_Handling
- State_Interval
- State_Repeat
- State_Time_Out
- State_Lock_Unlock
- State_Verify
- State_Release
- State_Destroy
- **Product Activate**
- Parent_State_Override
- Get_Parent_State
- Get_Child_State
- Remove_Child_State
- Get_State_ID
- Get_Running_State
- Get_Current_State
- Get_Locking_State
- Get_Elapsed_Time

 Delete HSMO

Hierarchical State Machine Object (HSMO)

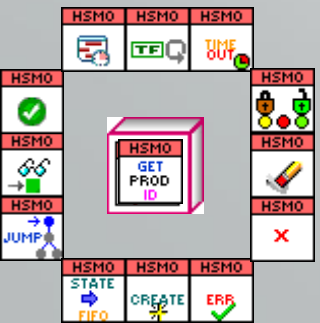
- ❖ Class Member (23)
- Get_Product_ID
 - State_Create
 - State_FIFO
 - State_LIFO
 - State_Acquisition
 - State_Error_Handling
 - State_Interval
 - State_Repeat
 - State_Time_Out
 - State_Lock_Unlock
 - State_Verify
 - State_Release
 - State_Destroy
 - Product_Activate
 - Parent State Override
 - Get_Parent_State
 - Get_Child_State
 - Remove_Child_State
 - Get_State_ID
 - Get_Running_State
 - Get_Current_State
 - Get_Locking_State
 - Get_Elapsed_Time



❖ Method :



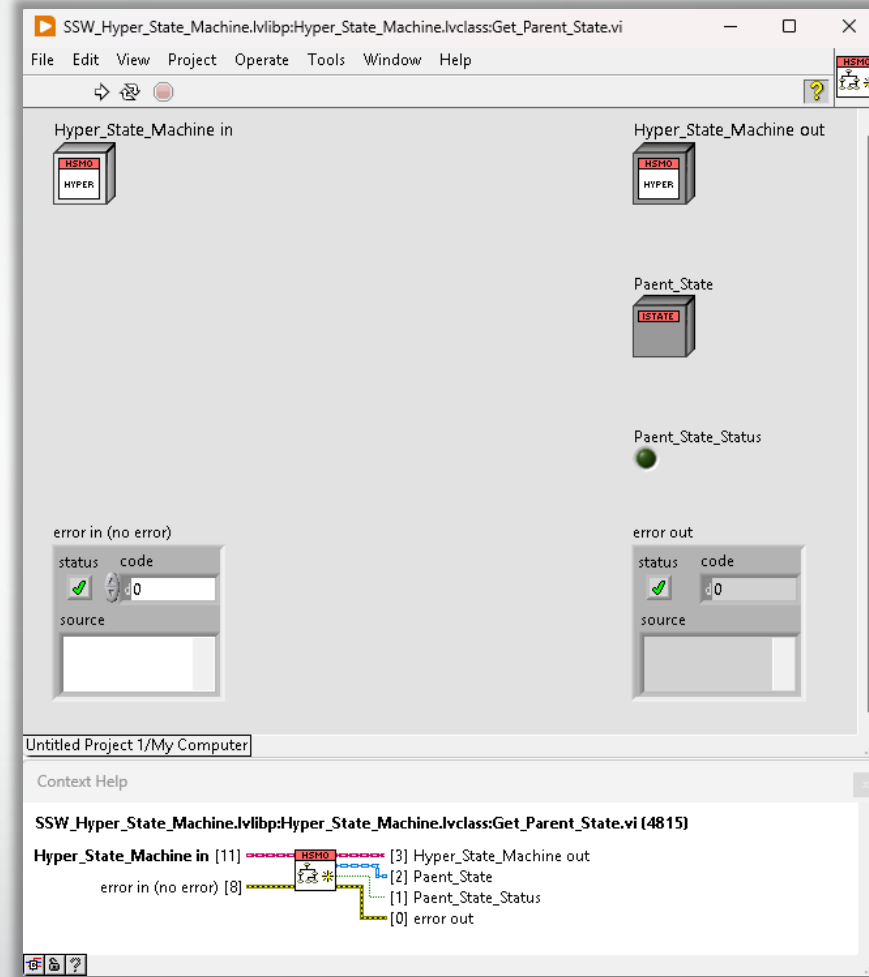
➤ Override the Original Parent State to Other Parent



Hierarchical State Machine Object (HSMO)

Class Member (23)

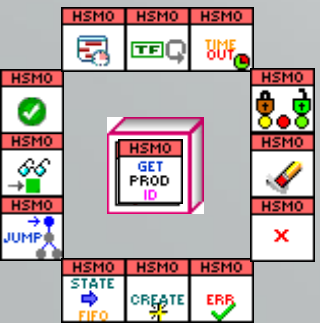
- Get_Product_ID
- State_Create
- State_FIFO
- State_LIFO
- State_Acquisition
- State_Error_Handling
- State_Interval
- State_Repeat
- State_Time_Out
- State_Lock_Unlock
- State_Verify
- State_Release
- State_Destroy
- Product_Activate
- Parent_State_Override
- Get_Parent_State
- Get_Child_State
- Remove_Child_State
- Get_State_ID
- Get_Running_State
- Get_Current_State
- Get_Locking_State
- Get_Elapsed_Time



Method :



Get Parent State of Current Child State

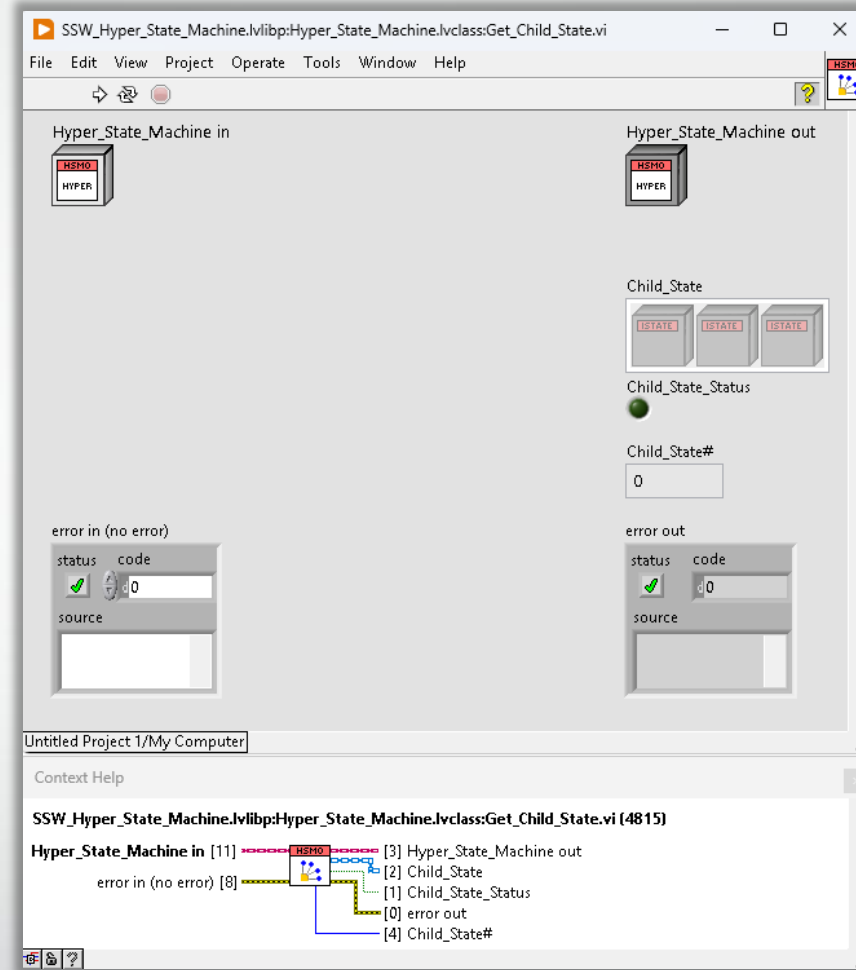


Hierarchical State Machine Object (HSMO)



Class Member (23)

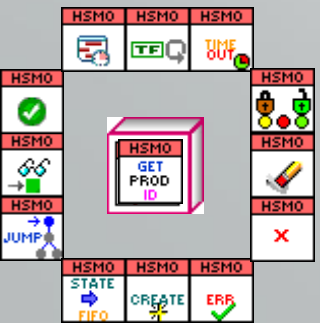
- Get_Product_ID
- State_Create
- State_FIFO
- State_LIFO
- State_Acquisition
- State_Error_Handling
- State_Interval
- State_Repeat
- State_Time_Out
- State_Lock_Unlock
- State_Verify
- State_Release
- State_Destroy
- Product_Activate
- Parent_State_Override
- Get_Parent_State
- Get_Child_State
- Remove_Child_State
- Get_State_ID
- Get_Running_State
- Get_Current_State
- Get_Locking_State
- Get_Elapsed_Time



Method :



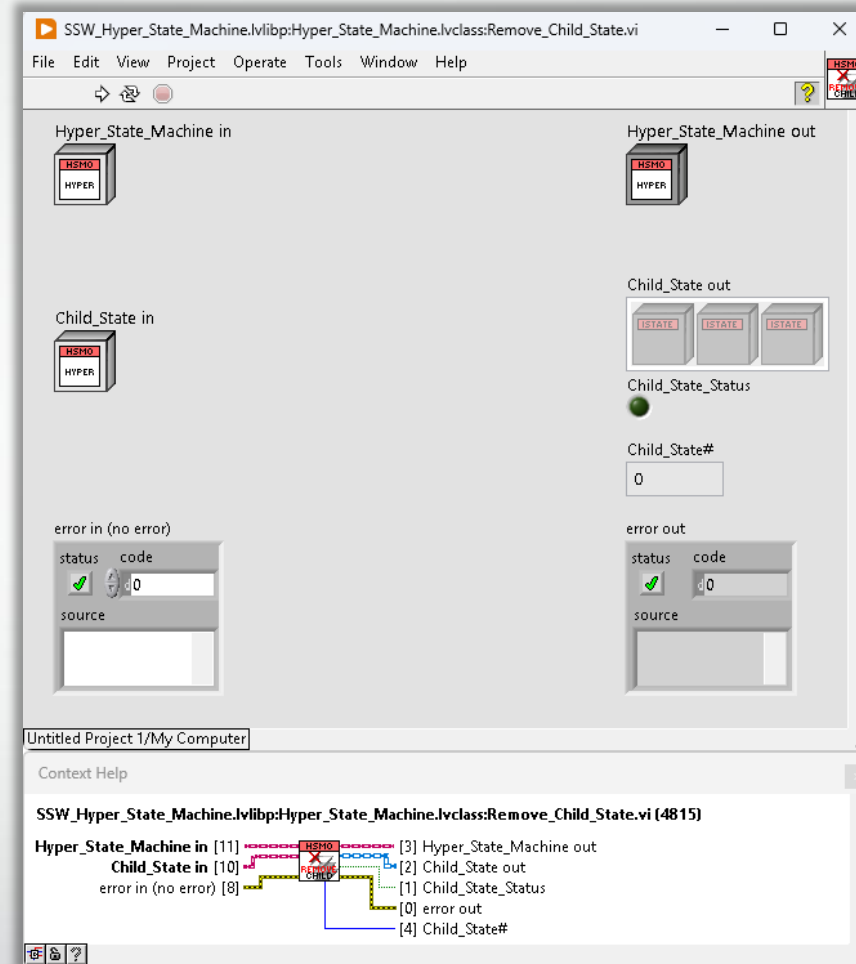
Get Child State Member of Current State (Parent)



Hierarchical State Machine Object (HSMO)

Class Member (23)

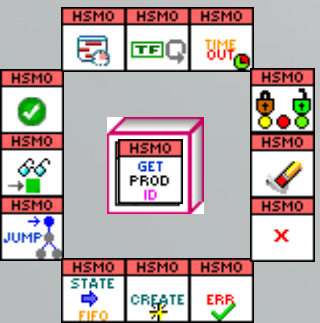
- Get_Product_ID
- State_Create
- State_FIFO
- State_LIFO
- State_Acquisition
- State_Error_Handling
- State_Interval
- State_Repeat
- State_Time_Out
- State_Lock_Unlock
- State_Verify
- State_Release
- State_Destroy
- Product_Activate
- Parent_State_Override
- Get_Parent_State
- Get_Child_State
- Remove_Child_State
- Get_State_ID
- Get_Running_State
- Get_Current_State
- Get_Locking_State
- Get_Elapsed_Time



Method :

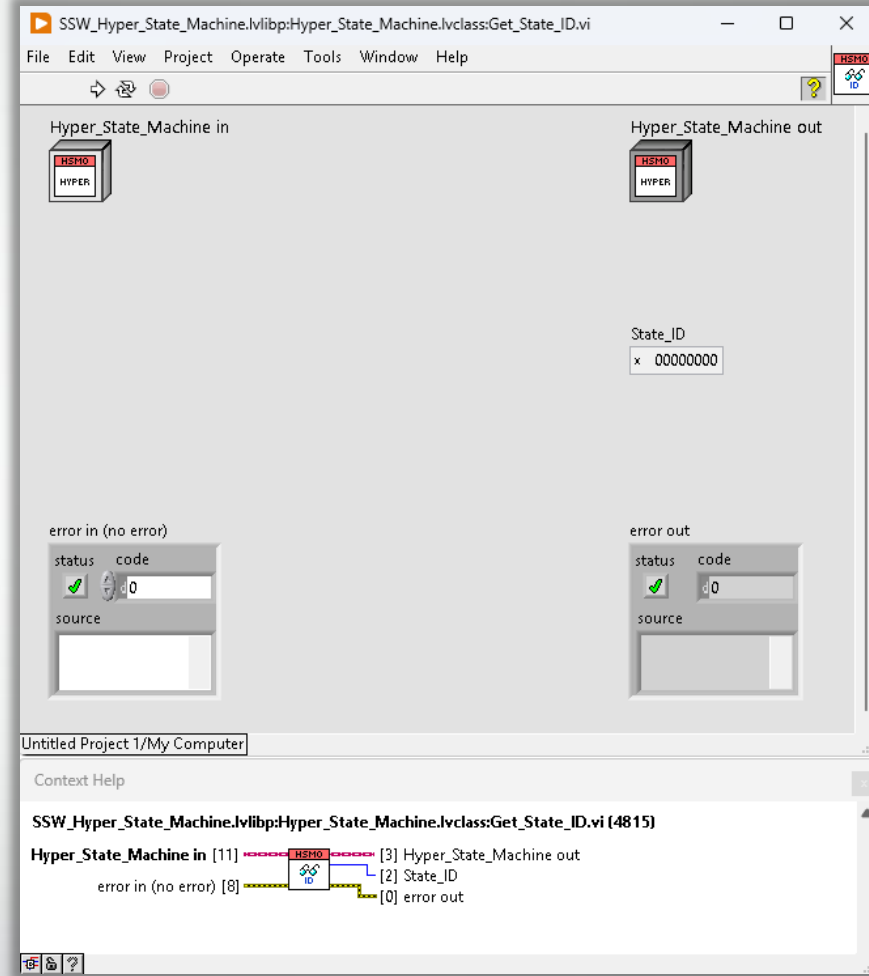


Remove Child State Member (*This State is only Internal use and Protected Scope*)

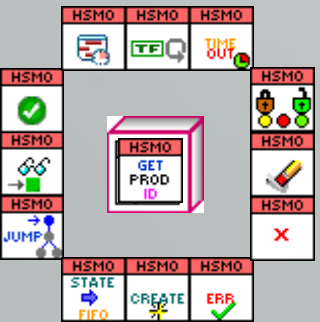


Hierarchical State Machine Object (HSMO)

- Get_Product_ID
- State_Create
- State_FIFO
- State_LIFO
- State_Acquisition
- State_Error_Handling
- State_Interval
- State_Repeat
- State_Time_Out
- State_Lock_Unlock
- State_Verify
- State_Release
- State_Destroy
- Product_Activate
- Parent_State_Override
- Get_Parent_State
- Get_Child_State
- Remove_Child_State
- [Get_State_ID](#)
- Get_Running_State
- Get_Current_State
- Get_Locking_State
- Get_Elapsed_Time



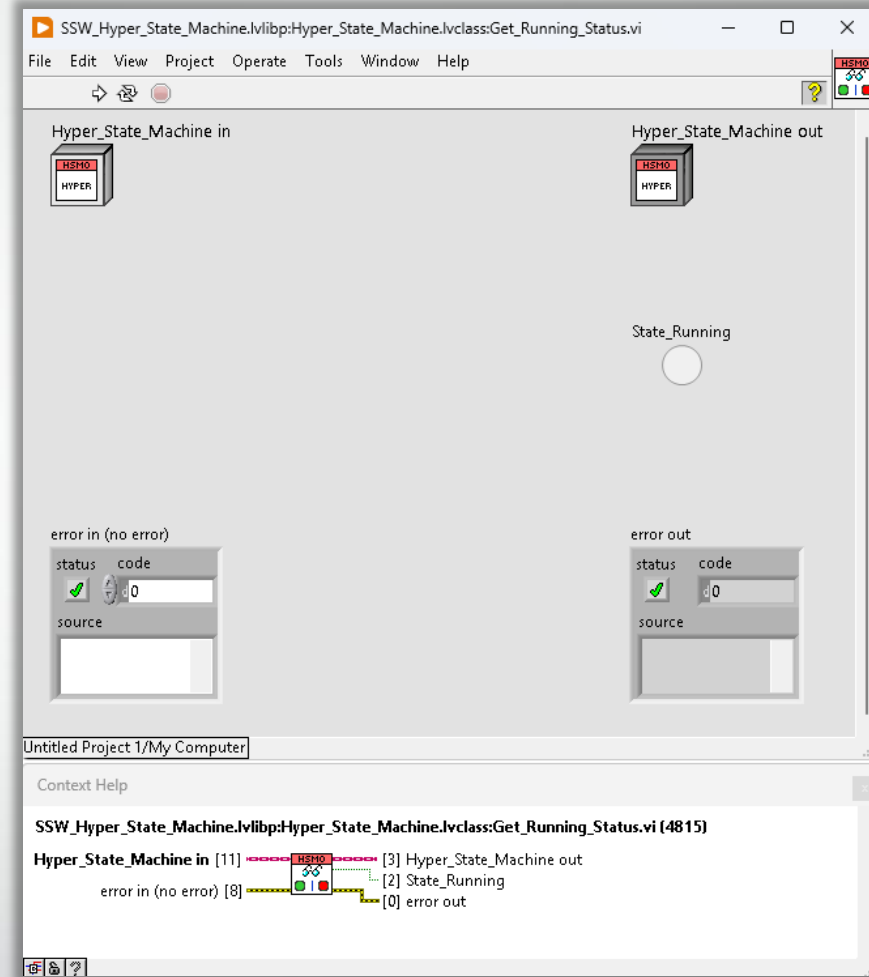
➤ Get Current State ID (Unique ID of Created State , This is Special Feature of DVR)



Hierarchical State Machine Object (HSMO)

Class Member (23)

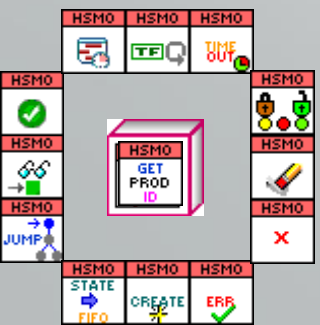
- Get_Product_ID
- State_Create
- State_FIFO
- State_LIFO
- State_Acquisition
- State_Error_Handling
- State_Interval
- State_Repeat
- State_Time_Out
- State_Lock_Unlock
- State_Verify
- State_Release
- State_Destroy
- Product_Activate
- Parent_State_Override
- Get_Parent_State
- Get_Child_State
- Remove_Child_State
- Get_State_ID
- Get_Running_State
- Get_Current_State
- Get_Locking_State
- Get_Elapsed_Time



Method :



Get Current State Status is Pending Run(True) or Stopped (False)

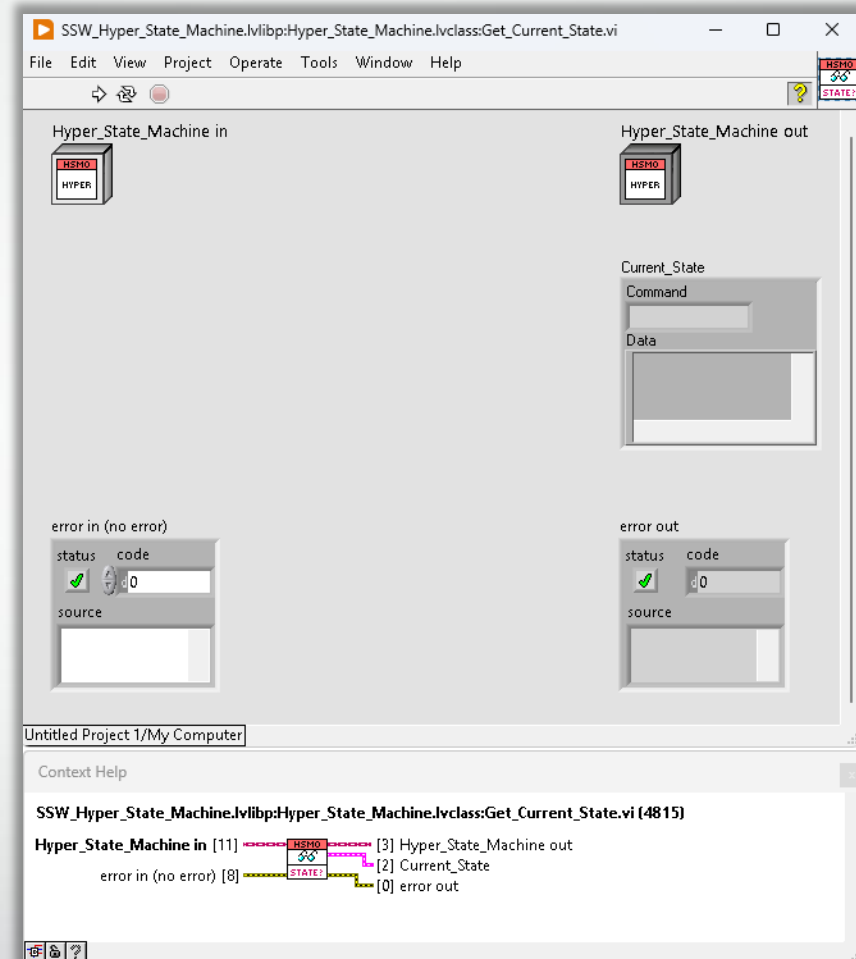


Hierarchical State Machine Object (HSMO)



Class Member (23)

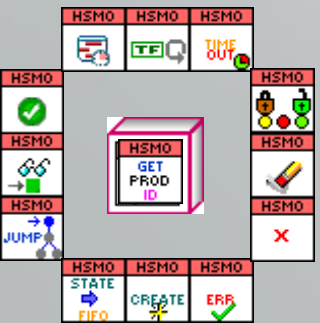
- Get_Product_ID
- State_Create
- State_FIFO
- State_LIFO
- State_Acquisition
- State_Error_Handling
- State_Interval
- State_Repeat
- State_Time_Out
- State_Lock_Unlock
- State_Verify
- State_Release
- State_Destroy
- Product_Activate
- Parent_State_Override
- Get_Parent_State
- Get_Child_State
- Remove_Child_State
- Get_State_ID
- Get_Running_State
- Get_Current_State
- Get_Locking_State
- Get_Elapsed_Time



Method :



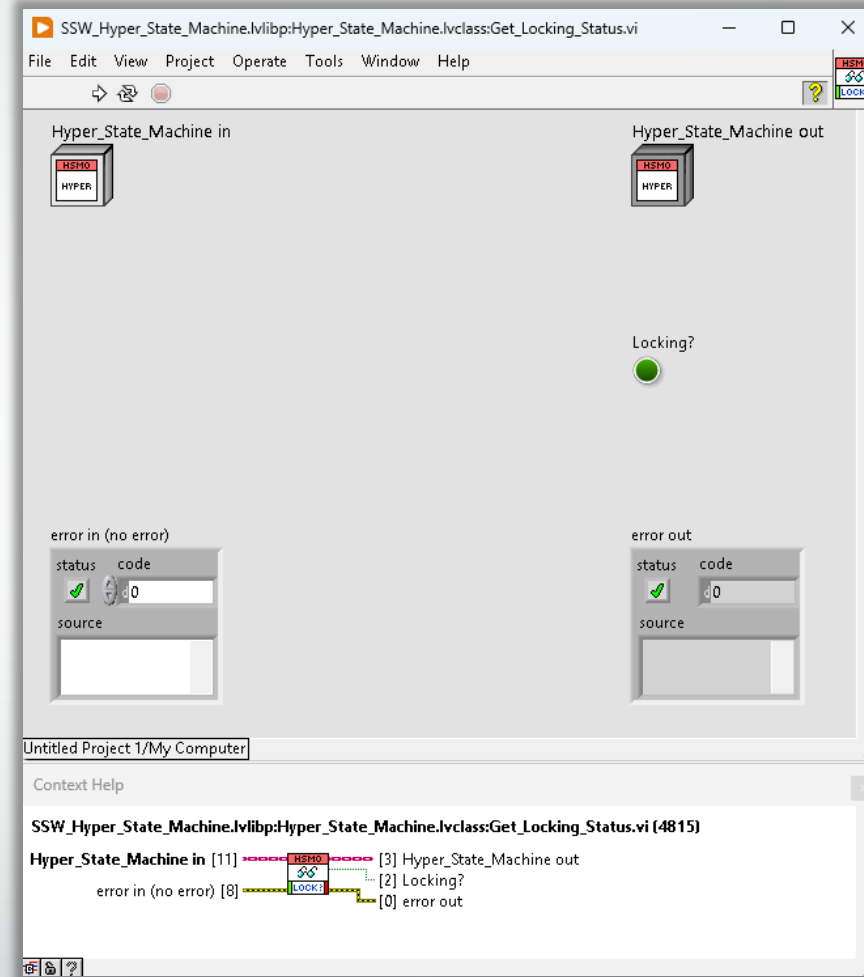
Monitoring the Current (do not remove out state)



Hierarchical State Machine Object (HSMO)

Class Member (23)

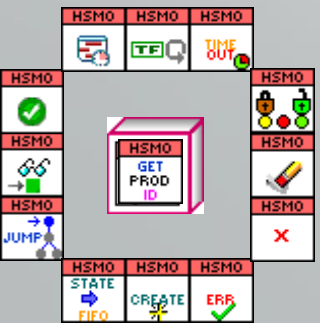
- Get_Product_ID
- State_Create
- State_FIFO
- State_LIFO
- State_Acquisition
- State_Error_Handling
- State_Interval
- State_Repeat
- State_Time_Out
- State_Lock_Unlock
- State_Verify
- State_Release
- State_Destroy
- Product_Activate
- Parent_State_Override
- Get_Parent_State
- Get_Child_State
- Remove_Child_State
- Get_State_ID
- Get_Running_State
- Get_Current_State
- Get_Locking_State
- Get_Elapsed_Time



Method :

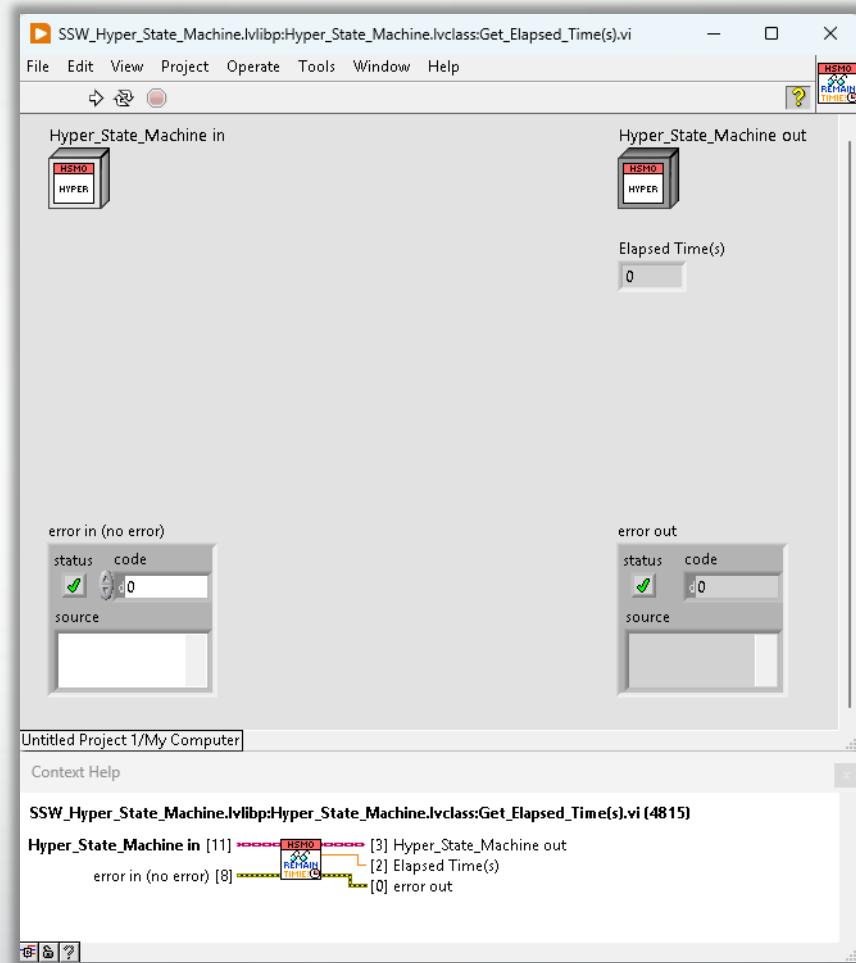


Get Current State is Locking or Free

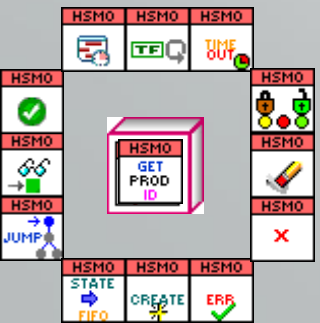


Hierarchical State Machine Object (HSMO)

- Get_Product_ID
- State_Create
- State_FIFO
- State_LIFO
- State_Acquisition
- State_Error_Handling
- State_Interval
- State_Repeat
- State_Time_Out
- State_Lock_Unlock
- State_Verify
- State_Release
- State_Destroy
- Product_Activate
- Parent_State_Override
- Get_Parent_State
- Get_Child_State
- Remove_Child_State
- Get_State_ID
- Get_Running_State
- Get_Current_State
- Get_Locking_State
- [Get_Elapsed_Time](#)



Get Current Elapsed Timing when State_Time_Out or State_Interval is Active



Hierarchical State Machine Object (HSMO)

Design Pattern



Object Create

Action

Acquisition

Execution

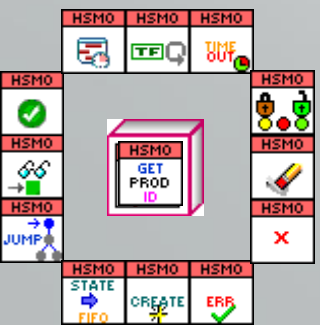
Error Handling

Object Destroy



Class Member

User Command

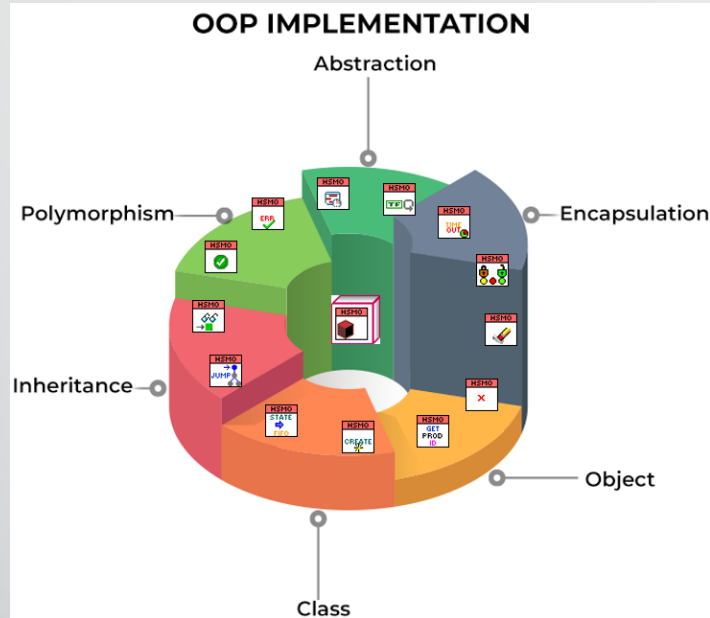


Implementation

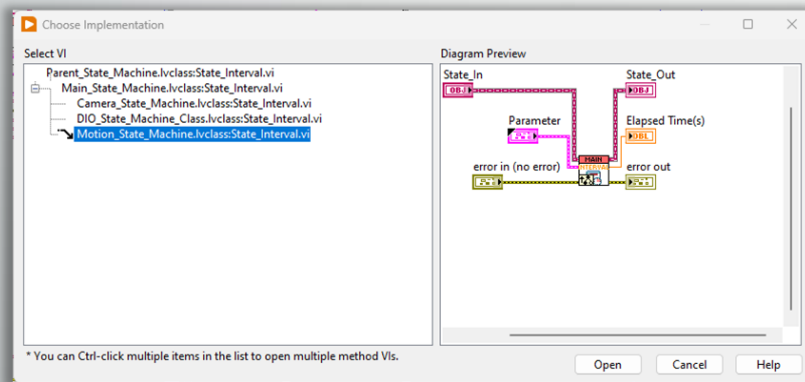


Hierarchical State Machine Object (HSMO)

❖ Benefit



Dynamic Dispatching Implementation



❖ Reduce Develop Time :

Code reusability, an important feature of Object-Oriented Programming (OOP), is enabled through inheritance, polymorphism, and information hiding. With inheritance, an object can be extended and code from the parent object can be reused or overloaded in the child object. For example we use one (Parent Class) and after that using Inheritance behavior form Parent Class to many Delivered Class and every property and Method of Parent Class is automatic transfer to Delivered class

Note : The Traditional Software Programming using Duplicate Code when we need more function and this is take time to debug or hard to maintain coding.

❖ Automatic Method Handling :

Dynamic Dispatching : Abstract parent class is automatic handle of many delivered class, with this behavior we able to Simulation Software without actual hardware by using Abstract Parent Class and switch to Actual hardware when ready (Hardware Abstraction)

❖ Low Resource and Fast Execution :

➤ This library is the Precompile code , it's low memory usage require and fast execution

❖ Multi Execution Rate Support :

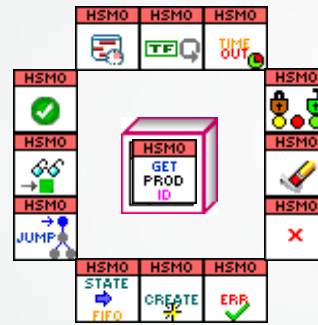
➤ All state task has the individual clock and mechanism managements, so we can mixed of low transfer rate and hi transfer rate device together.

❖ Asset Protection :

➤ This Framework is protected with standard license activate method , user can applied to protect own asset when machine or system is deliver to customer

❖ Large Scale Application Support :

OOP it's powerful and flexible of Programming Corroborate , Team Member is able to work with same Parent Class or Using Override Method to Make new Method that Inheritance from Parent Class with is not impact to other class and if Parent Class is has new method the Delivered Class is has new method also



Thank you